Symplectic Topology

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Based on the book "Introduction to Symplectic Topology" *by Dusa McDuff & Dietmar Salamon*

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I. Foundations

1 Symplectic Linear Algebra

1.1 Symplectic spaces

Definition 1.1.1

Let V be a finite-dimensional vector space. The **kernel** of a 2-form $\omega \in \Lambda^2(V^*)$ is the subspace

$$\ker \omega = \{v \in V \mid \omega(v, \bullet) = 0\}.$$

A 2-form is **nondegenerate** if its kernel is zero.
A nondegenerate 2-form is called a **symplectic form** and the pair (V, ω) a **symplectic vector space**.

Example 1.1.2

The archetypal symplectic vector space is $\mathbb{R}^{2n}$ equipped with the 2-form

$$\omega_0 = \sum_{i=1}^{n} dx_i \wedge dy_i$$

where $x_1, \dots, x_n, y_1, \dots, y_n$ forms the standard basis and dx_i, dy_i form its dual basis.

It is clear that a 2-form ω is nondegenerate iff its representation matrix relative to any basis is non-singular. The representation matrix relative to $B = (v_1, \dots, v_n)$ is $\omega_{ij} = \omega(v_i, v_j)$. Equivalently, if the induced map $\widehat{\omega}\colon V \to V^*$ given by $\widehat{\omega}(v)(u) = \omega(v, u)$ is injective, equivalently an isomorphism.

Definition 1.1.3

Let (V, ω_V) and (W, ω_W) be two symplectic spaces. A **symplectomorphism** is a linear isomorphism $\varphi\colon V \to W$ such that the pullback of ω_W is ω_V: $\omega_V = \varphi^*\omega_W$. Recall that the pullback is defined by

$$(\varphi^*\omega_W)(v, u) = \omega_W(\varphi v, \varphi u).$$

It is clear that the composition and inverse of symplectomorphisms are symplectomorphisms, so the category of symplectic spaces with symplectomorphisms forms a groupoid.
The collection of symplectomorphisms $V \to V$ is a group denoted by $\mathrm{Sp}(V, \omega)$.

Example 1.1.4

In the case of the standard symplectic structure on $\mathbb{R}^{2n}$, we use the notation $\mathrm{Sp}(2n) = \mathrm{Sp}(\mathbb{R}^{2n}, \omega_0)$. Notice that the representation of ω_0 relative to the standard basis is

$$J_0 = \begin{pmatrix} 0 & -I_n \\ I_n & 0 \end{pmatrix}.$$

Thus $\mathrm{Sp}(2n)$ can be viewed as the subgroup of $\mathrm{GL}_{2n}(\mathbb{R})$ defined by

$$\mathrm{Sp}(2n) = \left\{A \in \mathrm{GL}_{2n}(\mathbb{R}) \;\middle|\; A^\top J_0 A = J_0\right\}.$$

Proposition 1.1.5

$\mathrm{Sp}(2n)$ is a Lie group with Lie algebra

$$\mathfrak{sp}(2n) = \{-J_0 S \mid S \in \mathrm{Mat}_{2n} \text{ symmetric}\}.$$

Proof

Let $\mathfrak{so}(2n) \leq \mathrm{Mat}_{2n}$ be the linear subspace of skew-symmetric matrices, and define the map

$$f\colon \mathrm{Mat}_{2n} \to \mathfrak{so}(2n), \qquad f(A) = A^\top J_0 A.$$

Then $\mathrm{Sp}(2n) = f^{-1}J_0$, so to show that $\mathrm{Sp}(2n)$ is a Lie subalgebra of $\mathrm{GL}_{2n}(\mathbb{R})$, it is sufficient to show that J_0 is a regular value of f.

Note that f's differential at $A \in \mathrm{Mat}_{2n}$ is

$$df_A(X) = X^\top J_0 A + A^\top J_0 X.$$

If $f(A) = J_0$ and $B \in \mathfrak{so}(2n)$ (which is equal to $\mathfrak{so}(2n)$'s tangent space), then $X = -\frac{1}{2}AJ_0B$ is a preimage of B. Thus df_A is surjective so that J_0 is a regular value as desired.

Now,

$$\mathfrak{sp}(2n) = T_I \mathrm{Sp}(2n) = \ker df_I = \left\{X \in \mathrm{Mat}_{2n} \;\middle|\; X^\top J_0 + J_0 X = 0\right\}.$$

Note that if $X \in \mathfrak{sp}(2n)$ then $X = -J_0 S$ where $S = J_0 X$, and this is symmetric:

$$S^\top = -X^\top J_0 = J_0 X = S.$$

Conversely for S symmetric, $-J_0 S \in \mathfrak{sp}(2n)$. ∎

Definition 1.1.6

Let (V, ω) be a symplectic space and $W \leq V$ a subspace. Its **symplectic complement** is the subspace

$$W^\omega = \{v \in V \mid \omega(v, w) = 0 \text{ for all } w \in W\}.$$

Note that unlike the case for inner products, W^ω and W need not be transversal. The subspace W is called

(1) **isotropic** if $W \leq W^\omega$,

(2) **coisotropic** if $W^\omega \leq W$,

(3) **Lagrangian** if $W = W^\omega$,

(4) **symplectic** if $W \cap W^\omega = 0$.

Note that W is isotropic iff ω vanishes on W, and symplectic iff $(W, \omega|_W)$ is a symplectic space.

Lemma 1.1.7

For a subspace $W \leq V$,

$$\dim W + \dim W^\omega = \dim V, \qquad W^{\omega\omega} = W.$$

Proof

Consider $\widehat{\omega}\colon V \to V^*$ which is an isomorphism by non-degeneracy. Let us define

$$\operatorname{ann} W = \{\varphi \in V^* \mid \varphi(W) = 0\}.$$

Because $\widehat{\omega}$ is an isomorphism, we have

$$\operatorname{ann} W = \{\widehat{\omega}(v) \mid \widehat{\omega}(v)(W) = 0\} = \widehat{\omega}(W^\omega)$$

as desired.

Notice that a functional in $\operatorname{ann} W$ is the same as a functional over V/W. Indeed, $\varphi \in \operatorname{ann} W$ iff $W \leq \ker \varphi$ iff we can quotient φ by W. Thus $\operatorname{ann} W \cong (V/W)^*$ which has dimension $\dim V - \dim W$ as desired. ■

Let V be a $2n$-dimensional vector space with basis $u_1, \ldots, u_n, v_1, \ldots, v_n$. Then we can define a symplectic form on V by

$$\omega = \sum_{i=1}^{n} du_i \wedge dv_i,$$

equivalently

$$\omega(u_i, u_j) = \omega(v_i, v_j) = 0, \quad \omega(u_i, v_j) = \delta_{ij}, \qquad [\omega] = J_0.$$

Conversely any basis of a symplectic space which represents the symplectic form like this is called a **symplectic basis**.

Notice that the existence of a symplectic basis is equivalent to the existence of a symplectomorphism $(\mathbb{R}^{2n}, \omega_0) \to (V, \omega)$.

Theorem 1.1.8

Let (V, ω) be a symplectic space. Then V has even dimension, $\dim V = 2n$, and there exists a symplectic basis.

Proof

We induct on $m = \dim V$. For $m = 0$ this is trivial, so suppose $m > 0$. Because ω is non-degenerate there are $u_1, v_1 \in V$ such that $\omega(u_1, v_1) \neq 0$. We can normalize this and assume $\omega(u_1, v_1) = 1$.

Because ω is alternating, this means that u_1, v_1 are linearly independent. Furthermore the subspace W spanned by u_1, v_1 is symplectic as ω is nondegenerate there. Thus $V = W \oplus W^\omega$, and by induction W^ω has even dimension and a symplectic basis. By adding u_1, v_1 to W^ω's symplectic basis we get a symplectic basis of V, and V has even dimension. ■

An immediate corollary of this theorem is the following.

Corollary 1.1.9

Two symplectic spaces V, W are symplectomorphic iff $\dim V = \dim W$.

Corollary 1.1.10

Let V be a $2n$-dimensional vector space and ω a 2-form on V. Then ω is nondegenerate iff its n-fold exterior power ω^n is nonzero.

Proof

If ω is degenerate then there is a $v \neq 0$ such that $v \lrcorner \omega = 0$. Because interior multiplication is an antiderivation:

$$v \lrcorner \omega^n = \omega \wedge (v \lrcorner \omega^{n-1}) + (v \lrcorner \omega) \wedge \omega^{n-1} = \omega \wedge (v \lrcorner \omega^{n-1}).$$

Thus inductively we have $v \lrcorner \omega^n = 0$. Because ω^n is a top-form this means $\omega^n = 0$ (we can see this by extending v to a basis, then ω on that basis is zero).

Conversely if ω is nondegenerate, then by the previous theorem there is a symplectic basis

$$\omega = \sum_i dx_i \wedge dy_i.$$

Then

$$\omega^n = \sum_I dx_{i_1} \wedge dy_{i_1} \wedge \cdots \wedge dx_{i_n} \wedge dy_{i_n},$$

and we may remove Is with repeated indices, so the sum is over permutations of $1, \ldots, n$. Each permutation gives the same product as 2-forms commute, thus

$$\omega^n = n!\, dx_1 \wedge dy_1 \wedge \cdots dx_n \wedge dy_n \neq 0.$$ ■

Thus ω^n is an orientation form of V when ω is a symplectic form. In the proof above we showed that

$$\omega^n = n!\, dx_1 \wedge dy_1 \wedge \cdots dx_n \wedge dy_n,$$

which means that it corresponds to the oriented basis $(x_1, y_1, \ldots, x_n, y_n)$.

Proposition 1.1.11

The Lagrangian subspaces of V are precisely the maximal isotropic subspaces. Moreover, every basis of a Lagrangian subspace can be extended to a symplectic basis of V.

Proof

If W is isotropic then $W \leq W^\omega$ meaning $\dim W \leq \frac{1}{2} \dim V$. Since Lagrangian subspaces have dimension equal to $\frac{1}{2} \dim V$, Lagrangian subspaces are maximal. Conversely if W if isotropic and $W \neq W^\omega$, then there is a $v \in W^\omega - W$. Adding v to W results in an isotropic subspace as ω vanishes on $W + v$. Thus maximal isotropic subspaces must have $W = W^\omega$ so that W is Lagrangian.

For the second point it is sufficient to consider $V = \mathbb{R}^{2n}$ with the standard symplectic structure. Let $\Lambda \leq V$ be Lagrangian, then $\Lambda' = J_0 \Lambda$ is symplectic as well. Notice that $\Lambda' \cap \Lambda = 0$ because for $u \in \Lambda$ we have $\omega_0(u, J_0 u) = -u^\top u \neq 0$. Thus $\mathbb{R}^{2n} = \Lambda' \oplus \Lambda$.

Let $u_1, \ldots, u_n$ be a basis for Λ. Consider the map $\widehat{\omega} \colon \Lambda' \to \Lambda^*$. This is injective as its kernel is $\Lambda \cap \Lambda^\omega = 0$, and thus defines an isomorphism. Let $\varphi_1, \ldots, \varphi_n$ be the dual basis of $u_1, \ldots, u_n$, and thus there are $v_1, \ldots, v_n \in \Lambda'$ such that $\varphi_i = \widehat{\omega}(v_i)$.

Because Λ, Λ' are Lagrangian we have that

$$\omega(u_i, u_j) = \omega(v_i, v_j) = 0, \qquad \omega(v_i, u_j) = \varphi_i(u_j) = \delta_{ij}$$

as desired. ■

Proposition 1.1.12

Let $W \leq V$ be coisotropic.

(1) The quotient $\overline{W} = W/W^\omega$ has a unique symplectic structure $\overline{\omega}$ whose pullback under projection $W \to \overline{W}$ agrees with $\omega|_W$.

(2) If $\Lambda \leq V$ is Lagrangian, then

$$\overline{\Lambda} = \frac{(\Lambda \cap W) + W^\omega}{W^\omega} \leq \overline{W}$$

is Lagrangian.

Proof

Let $[w] \in \overline{W}$ denote the coset of w. Notice that $\overline{\omega}$ must satisfy

$$\omega(w, w') = \overline{\omega}([w], [w']),$$

which uniquely determines $\overline{\omega}$, showing uniqueness. Because W is coisotropic, W^ω is an isotropic subspace of W, and so this is well-defined: for $v, v' \in W^\omega$ and $w, w' \in W$,

$$\omega(w + v, w' + v') = \omega(w, w') + \omega(v, w') + \omega(w, v') + \omega(v, v') = \omega(w, w').$$

The middle two terms are zero because $v, v' \in W^\omega$, and $v \in W^\omega \leq W$ so the last term is zero. Furthermore $\overline{\omega}$ is nondegenerate as if $\overline{\omega}([w], \bullet) = 0$, then $w \in W^\omega$ so $[w] = 0$. So we have shown (1).

For (2) we show that

$$\widetilde{\Lambda} = (\Lambda \cap W) + W^\omega \leq V$$

is Lagrangian. Indeed,

$$\widetilde{\Lambda}^\omega = (\Lambda \cap W)^\omega \cap W = (\Lambda^\omega + W^\omega) \cap W = \Lambda \cap W + W^\omega = \widetilde{\Lambda}.$$

And $\widetilde{\Lambda}$ quotients to $\overline{\Lambda}$, and $\widetilde{\Lambda}^\omega$ quotients to $\overline{\Lambda}^\omega$, which are then equal. ■

1.2 The symplectic group

Recall that we have defined a groupoid whose objects are symplectic spaces, and whose isomorphisms are the symplectomorphisms. And corollary 1.1.9 says that the dimension of a symplectic space is its only significant invariant. Let us denote by

$$\mathrm{Sp}(V, \omega) = \{\Psi \in \mathrm{GL}(V) \mid \Psi^* \omega = \omega\}$$

the automorphism group of (V, ω), called the **symplectic group**.

By proposition 1.1.5, $\mathrm{Sp}(V, \omega)$ is a Lie group. Following this proposition, we have the following result.

Theorem 1.2.1

The symplectic group $\mathrm{Sp}(V, \omega)$ is a Lie group with Lie algebra

$$\mathfrak{sp}(V, \omega) = \{T \in \mathrm{end}(V) \mid \omega(A\bullet, \bullet) + \omega(\bullet, A\bullet) = 0\}.$$

We can understand this group by focusing on the case of $(\mathbb{R}^{2n}, \omega_0)$. Then $\mathrm{Sp}(2n) = \mathrm{Sp}(\mathbb{R}^{2n}, \omega_0)$ may be thought of as a Lie group of matrices $\Psi \in \mathrm{Mat}_{2n}$ satisfying $\Psi^\top J_0 \Psi = J_0$. Such matrices are called **symplectic**.

Lemma 1.2.2

Symplectic matrices have determinant one.

Proof

By corollary 1.1.10, ω_0^n is nonzero and thus a volume form on $\mathbb{R}^{2n}$. Furthermore a symplectic matrix Ψ satisfies $\Psi^*\omega_0 = \omega_0$ and thus $\Psi^*\omega_0^n = \omega_0^n$. Because this is a volume form, $\det \Psi = 1$. ■

We identify $\mathbb{R}^{2n}$ with $\mathbb{C}^n$ by identifying $(x, y) \in \mathbb{R}^{2n}$ with $x + iy \in \mathbb{C}^n$. Then J_0 corresponds to multiplication by i in $\mathbb{C}^n$. With this identification, $\mathrm{GL}_n(\mathbb{C})$ is the subgroup of $\mathrm{GL}_{2n}(\mathbb{R})$ consisting of matrices which commute with J_0:

$$\mathrm{GL}_n(\mathbb{C}) = \{A \in \mathrm{GL}_{2n}(\mathbb{R}) \mid J_0 A = A J_0\}.$$

In this identification, a matrix $X + iY \in \mathrm{Mat}_n(\mathbb{C})$ corresponds to the matrix

$$X + iY \in \mathrm{Mat}_n(\mathbb{C}) \leftrightarrow \begin{pmatrix} X & -Y \\ Y & X \end{pmatrix} \in \mathrm{Mat}_{2n})(\mathbb{R}).$$

Consider the determinant of these matrices. Letting

$$P = \begin{pmatrix} I & iI \\ I & -iI \end{pmatrix}, \qquad P^{-1} = \frac{1}{2}\begin{pmatrix} I & I \\ -iI & iI \end{pmatrix},$$

then

$$P \begin{pmatrix} X & -Y \\ Y & X \end{pmatrix} P^{-1} = \begin{pmatrix} X + iY & \\ & X - iY \end{pmatrix},$$

so that

$$\det \begin{pmatrix} X & -Y \\ Y & X \end{pmatrix} = \det(X + iY)\det(X - iY) = \det(X + iY) \cdot \overline{\det(X + iY)} = |\det(X + iY)|^2.$$

Notice how the conjugate transpose acts on $\mathrm{Mat}_n(\mathbb{C})$:

$$(X + iY)^\dagger = (X - iY)^\top = \begin{pmatrix} X^\top & Y^\top \\ -Y^\top & X^\top \end{pmatrix} = \begin{pmatrix} X & -Y \\ Y & X \end{pmatrix}^\top.$$

Thus $X + iY \in \mathrm{U}(n)$ iff

$$\begin{pmatrix} X & -Y \\ Y & X \end{pmatrix}^{-1} = \begin{pmatrix} X^\top & Y^\top \\ -Y^\top & X^\top \end{pmatrix},$$

which is iff

$$XX^\top + YY^\top = I, \qquad XY^\top = YX^\top.$$

Notice that

$$i(X + iY)^\dagger (X + iY) = i \iff (X + iY)^\dagger = (X + iY)^{-1} \iff X + iY \in \mathrm{U}(n).$$

Thus we see

$$\mathrm{U}(n) = \mathrm{GL}_n(\mathbb{C}) \cap \mathrm{Sp}(2n).$$

In fact we can extend this result.

Lemma 1.2.3

$$\mathrm{Sp}(2n) \cap \mathrm{O}(2n) = \mathrm{Sp}(2n) \cap \mathrm{GL}_n(\mathbb{C}) = \mathrm{O}(2n) \cap \mathrm{GL}_n(\mathbb{C}) = \mathrm{U}(n).$$

Proof

A matrix $\Psi \in \mathrm{GL}_{2n}(\mathbb{R})$ satisfies

$$\Psi \in \mathrm{GL}_n(\mathbb{C}) \iff \Psi J_0 = J_0 \Psi, \qquad \Psi \in \mathrm{Sp}(2n) \iff \Psi^\top J_0 \Psi = J_0, \qquad \Psi \in \mathrm{O}(2n) \iff \Psi^\top \Psi = I.$$

Note that any of these two conditions imply the third, so we have the first two equalities. By the above discussion, we also get the final. ■

For a linear operator Ψ we denote its spectrum by $\sigma(\Psi)$; this is its set of eigenvalues.

Lemma 1.2.4

Let $\Psi \in \mathrm{Sp}(2n)$, then $\lambda \in \sigma(\Psi)$ iff $\lambda^{-1} \in \sigma(\Psi)$, and the multiplicities of λ and λ^{-1} are equal. If ± 1 is an eigenvalue of Ψ then it has even multiplicity.

Finally, if $\lambda, \lambda' \in \sigma(\Psi)$ are two (not necessarily distinct) eigenvalues such that $\lambda\lambda' \neq 1$, then $\omega_0(z, z') = 0$ for any eigenvectors of eigenvalues λ, λ' respectively.

Proof

The first statement is due to the fact that Ψ^{-1} is similar to $\Psi^\top$:

$$\Psi^\top = J_0 \Psi^{-1} J_0^{-1},$$

Ψ and $\Psi^\top$ share the same eigenvalues (with the same multiplicities), and λ is an eigenvalue of Ψ iff λ^{-1} is of Ψ^{-1} (and the multiplicities agree).

Thus the total multiplicity of the eigenvalues not equal to ± 1 is even (as they come in pairs). The determinant of Ψ is the product of eigenvalues, as they come in pairs with their inverses this is equal to $(-1)^m$ where m is -1's multiplicity. By lemma 1.2.2 this is equal to 1 so m is even. Because Ψ has dimension $2n$, the multiplicity of 1 must be even as well.

The last statement is because

$$\omega_0(z, z') = \omega_0(\Psi z, \Psi z') = \lambda\lambda' \omega_0(z, z')$$

and since $\lambda\lambda' \neq 1$, $\omega_0(z, z') = 0$. ■

Suppose D is a diagonal matrix, $D = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$, then define

$$D^\alpha = \mathrm{diag}(\lambda_1^\alpha, \ldots, \lambda_n^\alpha)$$

for any real $\alpha \in \mathbb{R}$ where this is defined. Suppose $A \in \mathrm{Mat}_n(\mathbb{R})$ is a diagonalizable matrix with $A = P^{-1} D P$, D diagonal. Then define

$$A^\alpha = P^{-1} D^\alpha P$$

for real α where this is defined. Note that this agrees with A^n for n natural. It is easy to see that this is well-defined (i.e. not dependent on P, D).

If A^α exists and $Av = \lambda v$, then $A^\alpha v = \lambda^\alpha v$. This is clear for diagonal A, and extends to diagonalizable A by extending v to an eigenbasis and using that P.

Note that if A's eigenvalues are all positive, then this is defined for all α. In particular if S is symmetric and positive definite, then its powers exist for all real α.

Lemma 1.2.5

Exponentiation is continuous as a map

$$\mathrm{Sym}_+ \times \mathbb{R} \to \mathrm{Sym}_+, \qquad (S, \alpha) \mapsto S^\alpha$$

where $\mathrm{Sym}_+ \subseteq \mathrm{Mat}_n$ is the subspace of positive-definite symmetric matrices.

Proof

Let $S \in \mathrm{Sym}_+$, and let its minimum and maximum eigenvalues by $\lambda_{\min}, \lambda_{\max}$ respectively. Then we know for all vectors v,

$$\lambda_{\min} |v|^2 \leq v^\top S v \leq \lambda_{\max} |v|^2$$

and these bounds are tight. We claim that there is an open neighborhood $S \in U \subseteq \mathrm{Sym}_+$ and numbers $0 < m < M$ such that every $A \in U$ has eigenvalues bound by m, M.

Let $m = \lambda_{\min}/2$, then for every symmetric matrix A and v of unit norm:

$$v^\top A v = v^\top S v + v^\top (A - S) v \geq \lambda_{\min} - \left| v^\top (A - S) v \right| \geq \lambda_{\min} - |(A - S)v| \geq \lambda_{\min} - |A - S|$$

where $|A - S|$ denotes the operator norm. Thus if $|A - S| < m$ then the right-hand side is equal to $\lambda_{\min}/2$ and thus $v^\top A v \geq \lambda_{\min}/2$; so all of A's eigenvalues are bound from below by m. So all matrices in the ball of radius m around S have eigenvalues bound from below by m.

We can bound this from above too: for unit norm v

$$v^\top A v \leq |A| \leq |S| + |A - S|,$$

and so if we require $|A - S| < 1$ then

$$v^\top A v \leq |S| + 1 = \lambda_{\max} + 1,$$

so matrices in the ball of radius 1 around S have eigenvalues bound from above by $M = \lambda_{\max} + 1$. So the desired open set is

$$U = \{A \in \mathrm{Sym}_+ \mid |A - S| < 1, m\}.$$

Instead of considering exponentiating over arbitrary α, we show it is continuous for $a \leq \alpha \leq b$ for any arbitrary a, b; this is clearly sufficient. Consider the map $f : [m, M] \times [a, b] \to \mathbb{R}$ by $f(x, t) = t^x$, this is continuous as $m > 0$. By Stone-Weirestrass, because the codomain is compact, we can approximate f uniformly by a sequence of polynomials $p_k \in \mathbb{R}[x, t]$.

Now suppose $A \in U$, so $A = P^{-1} \operatorname{diag}(\lambda_1, \ldots, \lambda_n) P$, and because p_k are polynomial

$$p_k(x, A) = P^{-1} \operatorname{diag}(p_k(x, \lambda_1), \ldots, p_k(x, \lambda_n)) P.$$

Thus

$$A^\alpha - p_k(\alpha, A) = P^{-1} \operatorname{diag}(\ldots, \lambda_i - p_k(\alpha, \lambda_i), \ldots) P.$$

Because $A \in U$ the λ_i fall in $[m, M]$, and so under the operator norm

$$|A^\alpha - p_k(\alpha, A)| = \max_i |\lambda_i - p_k(\alpha, \lambda_i)| \leq \max_{t \in [m, M]} |t^\alpha - p_k(\alpha, t)|.$$

And so we see that the map $F : U \times [a, b] \to \mathrm{Sym}_+$, $F(A, \alpha) = A^\alpha$, satisfies for all A, α

$$|F(A, \alpha) - p_k(\alpha, A)| \leq \max_{t \in [m, M], \alpha \in [a, b]} |t^\alpha - p_k(\alpha, t)|,$$

so that $(\alpha, A) \mapsto p_k(\alpha, A)$ uniformly converges to F. Because p_k are all polynomial and thus continuous, so too is F. ∎

Lemma 1.2.6

If $S \in \mathrm{Sp}(2n)$ is symmetric and positive definite, then $P^\alpha \in \mathrm{Sp}(2n)$ for all $\alpha \in \mathbb{R}$.

Proof

As S is symmetric and positive definite we can decompose $\mathbb{R}^{2n}$ into eigenspaces

$$\mathbb{R}^{2n} = \bigoplus_{i=1}^{k} E_i, \qquad E_i = \ker(\lambda_i I - S),$$

for $\lambda_i > 0$ the eigenvalues. For $z, w \in \mathbb{R}^{2n}$ let their projections into E_i be denoted z_i, w_i respectively. Then because S is symplectic

$$\omega_0(z_i, w_j) = \omega_0(Pz_i, Pw_j) = \lambda_i\lambda_j\omega_0(z_i, w_j).$$

So either $\lambda_i\lambda_j = 1$ or $\omega_0(z_i, w_j) = 0$, and thus Then

$$\omega_0(P^\alpha z, P^\alpha w) = \sum_{i,j}^{k} (\lambda_i\lambda_j)^\alpha \omega_0(z_i, w_j) = \sum_{i,j}^{k} \omega_0(z_i, w_j) = \omega_0(z, w)$$

so that $P^\alpha \in \mathrm{Sp}(2n)$. ■

Proposition 1.2.7

(1) $\mathrm{U}(n)$ is a maximal compact subgroup of $\mathrm{Sp}(2n)$.

(2) The inclusion $\mathrm{U}(n) \to \mathrm{Sp}(2n)$ is a strong deformation retract. In particular $\mathrm{Sp}(2n)$ is connected.

Proof

We first prove (2). Define

$$H\colon [0,1] \times \mathrm{Sp}(2n) \to \mathrm{Sp}(2n), \qquad H(t, \Psi) = \Psi(\Psi^\top\Psi)^{-t/2}.$$

This is continuous by lemma 1.2.5, as $\Psi^\top\Psi$ is positive definite for any non-singular Ψ. This is well-defined by the previous lemma, $H(\Psi, t) \in \mathrm{Sp}(2n)$ for all t.

Now

$$H_0 = \mathrm{id}, \qquad H_t|_{\mathrm{U}(n)} = \mathrm{id}$$

where the latter is because $\Psi^\top\Psi = I$ for unitary Ψ. And for $\Psi \in \mathrm{Sp}(2n)$, $H_1(\Psi)$ is orthogonal:

$$H_1(\Psi)^\top H_1(\Psi) = (\Psi^\top\Psi)^{-1/2}\Psi^\top\Psi(\Psi^\top\Psi)^{-1/2} = I.$$

Because $\mathrm{Sp}(2n) \cap \mathrm{O}(2n) = \mathrm{U}(n)$, we have $H_1(\mathrm{Sp}(2n)) = \mathrm{U}(n)$. Thus H is indeed a strong deformation retract. And so $\mathrm{Sp}(2n)$ and $\mathrm{U}(n)$ are homotopy equivalent, and $\mathrm{U}(n)$ is connected so therefore $\mathrm{Sp}(2n)$ is as well.

We show (1) now. Suppose G is a compact subgroup of $\mathrm{Sp}(2n)$ containing $\mathrm{U}(n)$, but not equal to $\mathrm{U}(n)$. Then let $\Psi \in G - \mathrm{U}(n)$, and so $H_1(\Psi) = \Psi(\Psi^\top\Psi)^{-1/2} \in \mathrm{U}(n) \subseteq G$. Thus the positive definite symmetric matrix $P = \Psi^\top\Psi$ belongs to $G - \mathrm{U}(n)$. Therefore it must have an eigenvalue greater than 1: its eigenvalues come in pairs λ, λ^{-1}, so otherwise $P = I$. But then the sequence $P, P^2, P^3, \ldots$ in G has no convergent subsequence, contradicting G's compactness. ■

Remark 1.2.8

Every symplectic matrix $\Psi \in \mathrm{Sp}(2n)$ can be uniquely written as $\Psi = UP$ where $U \in \mathrm{U}(n)$ and P is a symmetric, positive definite, symplectic matrix. Existence is by

$$U = \Psi(\Psi^\top\Psi)^{-1/2}, \qquad P = (\Psi^\top\Psi)^{1/2}.$$

Uniqueness is because the only symmetric positive definite unitary matrix is I (because its eigenvalues come in pairs, and thus must all be 1). If $U \in \mathrm{U}(n)$ can be written as $U = U'P'$, then $P = U(U')^{-1}$ is unitary as well, and thus $P = I$ so $U = U'$. So this is true for unitary matrices, and thus if $UP = U'P'$ we have $U = U'P'P^{-1}$ so $U = U'$ and $P'P^{-1} = I$ meaning $P = P'$. This shows uniqueness.

This decomposition is called **symplectic polar decomposition**.

Proposition 1.2.9

The complex determinant map $\det_{\mathbb{C}}\colon \mathrm{U}(n) \to S^1$ induces an isomorphism of fundamental groups $\pi_1(\mathrm{U}\, n) \cong \pi_1(S^1) \cong \mathbb{Z}$.

Proof

The determinant map is a fibration with typical fiber $\mathrm{SU}(n)$. Indeed for every $e^{i\theta} \in S^1$ we can consider the open subset W of S^1 consisting of numbers whose arguments are between $\theta - \varepsilon$ and $\theta + \varepsilon$. Then $\det_{\mathbb{C}}^{-1}(W) \to \mathrm{SU}(n) \times W$ given by $\Psi \mapsto (e^{i\varphi/n}\Psi, e^{i\varphi})$ where $e^{i\varphi} = \det \Psi$ is a homeomorphism with inverse $(\Phi, e^{i\varphi}) \to e^{i\varphi/n}\Phi$. It also commutes with $\det_{\mathbb{C}}$ because $\det(e^{i\varphi/n}\Phi) = e^{i\varphi}$.

Thus there is an exact sequence

$$\pi_1(\mathrm{SU}(n)) \to \pi_1(\mathrm{U}\, n) \to \pi_1(S^1) \to \pi_0(\mathrm{SU}(n)).$$

It is sufficient then to show that $\mathrm{SU}(n)$ is simply connected, and we get the desired result.

It is trivial that $\mathrm{SU}(n)$ is simply connected for $n = 1$, as $\mathrm{SU}(1) = \{1\}$. For $n \geq 2$ the map $\mathrm{SU}(n) \to S^{2n-1}$ which maps $U \in \mathrm{SU}(n)$ to its first column is a fibration with typical fiber $\mathrm{SU}(n-1)$. And so we have an exact sequence

$$\pi_1(\mathrm{SU}(n-1)) \to \pi_1(\mathrm{SU}\, n) \to \pi_1(S^{2n-1}) \to \pi_0(\mathrm{SU}(n-1)).$$

By induction this means $\pi_1(\mathrm{SU}\, n) \cong \pi_1(S^{2n-1}) = 1$ because $2n - 1 > 1$. ■

Because $\mathrm{U}(n)$ is a deformation retract of $\mathrm{Sp}(2n)$, they have the same fundamental group so $\pi_1(\mathrm{Sp}(2n)) \cong \mathbb{Z}$. We now construct an explicit isomorphism.

Recall that the **degree** of a loop $\gamma\colon \mathbb{R}/\mathbb{Z} \to S^1$ is defined to be its value in $\mathbb{Z}$ under the usual isomorphism $\pi_1(S^1) \cong \mathbb{Z}$. Equivalently, we lift γ to $\widetilde{\gamma}\colon [0,1] \to \mathbb{R}$ (because covering spaces have the path lifting property), and define

$$\deg \gamma = \widetilde{\gamma}(1) - \widetilde{\gamma}(0).$$

Theorem 1.2.10

There exists a unique function μ, called the **Maslov index**, which assigns an integer $\mu(\Psi) \in \mathbb{Z}$ to every loop $\Psi\colon \mathbb{R}/\mathbb{Z} \to \mathrm{Sp}(2n)$ and satisfies the following axioms:

(1) Homotopy: two loops are homotopic iff they have the same Maslov index.

(2) Product: the Maslov index of the product of two loops is the sum of their Maslov indices:

$$\mu(\Psi_1\Psi_2) = \mu(\Psi_1) + \mu(\Psi_2).$$

Note here we are taking the pointwise product of loops of matrices: $(\Psi_1\Psi_2)(t) = \Psi_1(t)\Psi_2(t)$.

(3) Direct sum: if $n = n' + n''$, we may identify $\mathrm{Sp}(2n') \oplus \mathrm{Sp}(2n'')$ as a subgroup of $\mathrm{Sp}(2n)$. Then for loops Ψ', Ψ'' in $\mathrm{Sp}(2n'), \mathrm{Sp}(2n'')$ respectively,

$$\mu(\Psi' \oplus \Psi'') = \mu(\Psi') + \mu(\Psi'').$$

(4) Normalization: the loop $\Psi\colon \mathbb{R}/\mathbb{Z} \to \mathrm{Sp}(2)$ defined by

$$\Psi(t) = \begin{pmatrix} \cos(2\pi t) & -\sin(2\pi t) \\ \sin(2\pi t) & \cos(2\pi t) \end{pmatrix}$$

has Maslov index 1.

Note:
We identify $\mathrm{Sp}(2n') \oplus \mathrm{Sp}(2n'')$ with a subgroup of $\mathrm{Sp}(2n)$ by

$$\begin{pmatrix} X & Y \\ Z & W \end{pmatrix} \oplus \begin{pmatrix} X' & Y' \\ Z' & W' \end{pmatrix} = \left(\begin{array}{cc|cc} X & & Y & \\ & X' & & Y' \\ \hline Z & & W & \\ & Z & & W' \end{array}\right).$$

Proof

Define $\rho\colon \mathrm{Sp}(2n) \to S^1$ by mapping $\Psi \in \mathrm{Sp}(2n)$ (not a loop) to

$$\rho(\Psi) = \det_{\mathbb{C}}(X + iY), \qquad \begin{pmatrix} X & -Y \\ Y & X \end{pmatrix} = \Psi(\Psi^\top \Psi)^{-1/2} \in \mathrm{U}(n).$$

Then define the Maslov index to be

$$\mu(\Psi) = \deg(\rho \circ \Psi).$$

Equivalently $\mu(\Psi) = \alpha(1) - \alpha(0)$ where $\alpha\colon I \to \mathbb{R}$ is a continuous lift of $\rho \circ \Psi$, meaning

$$\exp(2\pi i \alpha(t)) = \det_{\mathbb{C}}(X(t) + iY(t)).$$

Clearly this definition depends only on the homotopy class of Ψ, and by above we know $\rho\colon \mathrm{Sp}(2n) \to S^1$ defines an isomorphism of the fundamental groups. This isomorphism is μ, so we have axiom (1). The product axiom is true for loops of unitary matrices, because $\rho(\Psi) = \det_{\mathbb{C}} \Psi$, so if α_i is a lift of $\rho\Psi_i$ then

$$\det_{\mathbb{C}}(\Psi_1\Psi_2) = \det_{\mathbb{C}} \Psi_1 \cdot \det_{\mathbb{C}} \Psi_2 = \exp(2\pi i \alpha_1(t)) \exp(2\pi i \alpha_2(t)) = \exp(2\pi i(\alpha_1 + \alpha_2)(t)).$$

Thus $\alpha_1 + \alpha_2$ is a lift of $\rho(\Psi_1\Psi_2)$, so $\mu(\Psi_1\Psi_2) = \mu(\Psi_1) + \mu(\Psi_2)$. Because a loop of symplectic matrices is homotopic to a loop of unitary matrices, we have the desired result.

The last two axioms are easily verified. Thus the existence is seen, and uniqueness can be shown as well. ■

1.3 Lagrangian subspaces

For a symplectic space (V, ω), let $\mathcal{L}(V, \omega)$ denote the set of its Lagrangian subspaces. We denote

$$\mathcal{L}(n) = \mathcal{L}(\mathbb{R}^{2n}, \omega_0).$$

We can characterize the Lagrangian subspaces of $\mathbb{R}^{2n}$ more explicitly. Firstly a Lagrangian subspace of $\mathbb{R}^{2n}$ has dimension n, and is thus the image of some full-rank linear transform $\mathbb{R}^n \to \mathbb{R}^{2n}$, or equivalently a $2n \times n$-matrix of rank n. We can then determine when such an image is Lagrangian.

Proposition 1.3.1

Let $X, Y \in \mathrm{Mat}_n$ be matrices, and define $\Lambda \leq \mathbb{R}^{2n}$ by

$$\Lambda = \operatorname{im} Z, \qquad Z = \begin{pmatrix} X \\ Y \end{pmatrix}.$$

Then $\Lambda \in \mathcal{L}(n)$ iff Z has rank n and $X^\top Y = Y^\top X$.
In particular the graph $\Lambda = \{(x, Ax) \mid x \in \mathbb{R}^n\}$ of some matrix $A \in \mathrm{Mat}_n$ is Lagrangian iff A is symmetric.

Proof

It is sufficient to consider when ω_0 is trivial on Λ. Let $z = (Xu, Yu)$ and $z' = (Xu', Yu')$ in Λ, then

$$\omega_0(z, z') = \begin{pmatrix} (Xu)^\top & (Yu)^\top \end{pmatrix} \begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix} \begin{pmatrix} Xu' \\ Yu' \end{pmatrix} = u^\top (X^\top Y - Y^\top X) u'.$$

This is zero for all u, u' iff $X^\top Y = Y^\top X$, as desired.
The second claim is because $\Lambda = \operatorname{im}(I, A)$, which has rank n. Thus it is Lagrangian iff $A^\top = A$. ■

A matrix $Z \in \mathbb{R}^{2n \times n}$ of the form

$$Z = \begin{pmatrix} X \\ Y \end{pmatrix}, \qquad X^\top Y = Y^\top X$$

is called a **Lagrangian frame**. Notice that Z's columns form an orthonormal basis for $\Lambda = \operatorname{im} Z$ iff $U = X + iY$ is unitary. Indeed Z's columns are orthonormal iff

$$I = Z^\top Z = X^\top X + Y^\top Y.$$

And U is unitary iff

$$I = U^\dagger U = (X^\top - iY^\top)(X + iY) = X^\top X + Y^\top Y + i(X^\top Y - Y^\top X) = X^\top X + Y^\top Y.$$

A Lagrangian frame Z for which $U = X + iY$ is unitary is called a **unitary Lagrangian frame**. Every Lagrangian subspace has a unitary Lagrangian frame as we can take an orthonormal basis and put it in a matrix.
Let us define the Lagrangian subspace

$$\Lambda_{\mathsf{hor}} = \left\{(x, 0) \in \mathbb{R}^{2n} \;\middle|\; x \in \mathbb{R}^n\right\} = \operatorname{im} \begin{pmatrix} I \\ 0 \end{pmatrix}.$$

Lemma 1.3.2

(1) If $\Lambda \in \mathcal{L}(n)$ and $\Psi \in \mathrm{Sp}(2n)$, then $\Psi\Lambda \in \mathcal{L}(n)$.

(2) For any two Lagrangian subspaces $\Lambda, \Lambda' \in \mathcal{L}(n)$, there is a symplectic matrix $\Psi \in \mathrm{U}(n)$ such that $\Lambda' = \Psi\Lambda$.

(3) $\mathcal{L}(n)$ can be seen to be naturally isomorphic to $\mathrm{U}(n)/\mathrm{O}(n)$, and thus carries the natural structure of a manifold of dimension $n(n+1)/2$.

Proof

(1) is clear. For (2) let $\Lambda \leq \mathbb{R}^{2n}$ be given by a unitary Lagrangian frame $Z = (X, Y)$ and define

$$\Psi = X + iY = \begin{pmatrix} X & -Y \\ Y & X \end{pmatrix} \in \mathrm{Sp}(2n) \cap \mathrm{O}(n).$$

Then

$$\Psi\Lambda_{\mathsf{hor}} = \Lambda,$$

and thus we have shown (2).

A unitary Lagrangian frame of Λ is uniquely determined up to choice of orthonormal basis. Altering the basis amounts to multiplication on the right by an orthonormal matrix, so we see that there is a bijection $\mathrm{U}(n)/\mathrm{O}(n) \cong \mathcal{L}(n)$. The dimension of $\mathrm{U}(n)$ is n^2, and the dimension of $\mathrm{O}(n)$ is $n(n-1)/2$, so $\dim \mathcal{L}(n) = n(n+1)/2$. ■

We can show that $\mathcal{L}(n) \cong \mathrm{U}(n)/\mathrm{O}(n)$ has a fundamental group isomorphic to the integers. This isomorphism is also called the **Maslov index**.

Theorem 1.3.3

There exists a unique function, the **Maslov index**, μ which assigns to each loop $\Lambda\colon \mathbb{R}/\mathbb{Z} \to \mathcal{L}(n)$ an integer and satisfies the following axions:

(1) homotopy: two loops in $\mathcal{L}(n)$ are homotopic iff they have the same Maslov index.

(2) product: for any two loops $\Lambda\colon \mathbb{R}/\mathbb{Z} \to \mathcal{L}(n)$ and $\Psi\colon \mathbb{R}/\mathbb{Z} \to \mathrm{Sp}(2n)$, we have

$$\mu(\Psi\Lambda) = \mu(\Lambda) + 2\mu(\Psi).$$

(3) direct sum: if $n = n' + n''$, then we can identify $\mathcal{L}(n') \oplus \mathcal{L}(n'')$ as a submanifold of $\mathcal{L}(n)$ and then

$$\mu(\Lambda' \oplus \Lambda'') = \mu(\Lambda') + \mu(\Lambda'').$$

(4) zero: a constant loop $\Lambda(t) = \Lambda_0$ has Maslov index zero.

(5) normalization: the loop $\Lambda\colon \mathbb{R}/\mathbb{Z} \to \mathcal{L}(1)$ given by

$$\Lambda(t) = e^{\pi i t}\mathbb{R} \leq \mathbb{C} = \mathbb{R}^2$$

has Maslov index one.

Proof

An orthonormal matrix has determinant ± 1, thus the map

$$\rho\colon \mathcal{L}(n) \cong \mathrm{U}(n)/\mathrm{O}(n) \to S^1, \qquad \rho([U]) = \det_{\mathbb{C}} U^2$$

is well-defined. Here $[U] \in \mathrm{U}(n)/\mathrm{O}(n)$ denotes the coset of the unitary matrix $U = X + iY$, which in turn represents the Lagrangian subspace $\Lambda = \mathrm{im}(X, Y)$.

Define the Maslov index to be the induced map $\pi_1(\mathcal{L}(n)) \to \pi_1(S^1) \xrightarrow{\sim} \mathbb{Z}$. That is for a loop Λ,

$$\mu(\Lambda) = \deg(\rho \circ \Lambda) = \alpha(1) - \alpha(0)$$

where $\alpha\colon I \to \mathbb{R}$ is a lift of $\rho \circ \Lambda$, meaning

$$e^{\pi i \alpha(t)} = \det_{\mathbb{C}}(U(t)), \qquad \Lambda = [U].$$

The Maslov index then depends only on the homotopy class of Λ. We show it satisfies the product axiom: first note that if Ψ is unitary $\rho(\Psi\Lambda) = \rho(\Psi)^2\rho(\Lambda)$ ($\rho(\Psi)$ corresponds to the ρ in theorem 1.2.10). And so the degree of this is

$$\deg(\rho(\Psi\Lambda)) = \deg \rho(\Lambda) + 2\deg \rho(\Psi) = \mu(\Lambda) + 2\mu(\Psi).$$

If Ψ is not unitary, it is homotopic to a loop of unitary matrices, and the result follows.

We now show the converse direction in the homotopy axiom. Suppose that Λ_0, Λ_1 are two loops of Lagrangian subspaces with the same Maslov index, we must show Λ_0, Λ_1 are homotopic. By the previous lemma and the product axiom, we may assume $\Lambda_j(0) = \Lambda_j(1) = \Lambda_{\mathsf{hor}}$. Lift $\Lambda_j(t)$ to $U_j(t) = X_j(t) + iY_j(t) \in \mathrm{U}(n)$ such that $U_0(0) = U_1(0) = I$ (as these represent Λ_{hor}). We may multiply U_j by right multiplication with orthogonal matrices to get $U_0(1) = U_1(1) = I$ as well. Then $U(t) = U_1(t)U_0(t)^{-1}$ forms a loop of unitary matrices.

Note that $\mu(\Lambda_0) = \mu(\Lambda_1)$ means $2\mu(U_0) = 2\mu(U_1)$, and thus U_0, U_1 are homotopic. Thus Λ_0, Λ_1 are homotopic.

The normalization axiom follows from the other axioms. Note that the loop $\Lambda(t) = e^{\pi i t}\mathbb{R}$ satisfies

$$\Lambda(t) \oplus \Lambda(t) = \Psi(t)\Lambda_0$$

where $\Lambda_0 = \mathbb{R}^2 \leq \mathbb{C}^2$ is constant and Ψ is the loop

$$\Psi(t) = e^{2\pi i t}X(t) = e^{\pi i t}\begin{pmatrix} \cos(\pi t) & -\sin(\pi t) \\ \sin(\pi t) & \cos(\pi t) \end{pmatrix}.$$

Here $X(t)$ has constant determinant 1 and so $\det \Psi = e^{2\pi i t}$ lifts to $\alpha(t) = t$ and thus $\mu(\Psi) = 1$. Thus we have

$$2\mu(\Lambda) = \mu(\Lambda \oplus \Lambda) = \mu(\Lambda_0) + 2\mu(\Psi) = 2 \Longrightarrow \mu(\Lambda) = 1. \qquad \blacksquare$$

1.4 The affine nonsqueezing theorem

Define an **affine symplectomorphism** of $\mathbb{R}^{2n}$ to be a map $\psi\colon \mathbb{R}^{2n} \to \mathbb{R}^{2n}$ of the form

$$\psi(z) = \Psi(z) + z_0, \qquad \Psi \in \mathrm{Sp}(2n),\ z_0 \in \mathbb{R}^{2n}.$$

Let $\mathrm{ASp}(2n)$ denote the set of affine symplectomorphisms, this is a group as

$$\psi \circ \psi'(z) = \Psi \circ \Psi'(z) + \Psi(z_0') + z_0, \qquad \psi^{-1}(z) = \Psi^{-1}(z) - \Psi^{-1}(z_0)$$

are affine symplectomorphisms.

The affine nonsqueezing theorem says that a ball in $\mathbb{R}^{2n}$ can only be embedded into a symplectic cylinder if the ball's radius is smaller than the cylinder. If we let $e_1, \ldots, e_n, f_1, \ldots, f_n$ denote the standard basis of $\mathbb{R}^{2n}$, then the symplectic cylinder of radius $R > 0$ is

$$Z^{2n}(R) = B^2(R) \times \mathbb{R}^{2n-2} = \left\{ z \in \mathbb{R}^{2n} \;\middle|\; \langle z, e_1\rangle^2 + \langle z, f_1\rangle^2 \leq R^2 \right\} \subseteq \mathbb{R}^{2n}.$$

Theorem 1.4.1 (The affine nonsqueezing theorem)

Let $\psi \in \mathrm{ASp}(2n)$ be an affine symplectomorphism such that $\psi(B^{2n}(r)) \subseteq Z^{2n}(R)$. Then $r \leq R$.

Proof

We may assume that $r = 1$ and let $\psi(z) = \Psi(z) + z_0$. Then define

$$u = \Psi^\top e_1, \qquad v = \Psi^\top f_1, \qquad a = \langle e_1, z_0\rangle, \qquad b = \langle f_1, z_0\rangle.$$

Because $\Psi \in \mathrm{Sp}(2n)$ we have

$$\omega_0(u, v) = \omega_0(\Psi^\top e_1, \Psi^\top f_1) = \omega_0(e_1, f_1) = 1.$$

The condition $\psi(B^{2n}(1)) \subseteq Z^{2n}(R)$ is equivalent to

$$\sup_{|z|=1}\left(\langle e_1, \psi(z)\rangle^2 + \langle f_1, \psi(z)\rangle^2\right) = \sup_{|z|=1}\left((\langle u, z\rangle + a)^2 + (\langle v, z\rangle + b)^2\right) \leq R^2.$$

Because $1 = \omega_0(u,v) \leq |u|\,|v|$ (as $\omega_0(x,y) = \langle x, J_0 y\rangle$ and J_0 is norm-preserving), one of u, v must have norm at least 1. Without loss of generality $|u| \geq 1$, and then let $z = \pm u/|u|$ whose sign agrees with a's. Then

$$(\langle u, z\rangle + a)^2 \geq |u|^2 \geq 1,$$

which then implies that $1 \leq R^2$ and so $1 \leq R$ as desired. ■

Because $B^{2n}(R) \subseteq Z^{2n}(R)$, we have that

$$R = \sup\left\{r > 0 \;\middle|\; \text{there exists an affine/linear symplectomorphism } \psi \text{ such that } \psi(B^{2n}(r)) \subseteq Z^{2n}(R)\right\}.$$

Either affine or linear in the above supremum gives the same result.

Call a set $B \subseteq \mathbb{R}^{2n}$ an **linear** (resp. **affine**) **symplectic ball** of radius r if it is linearly (resp. affinely) symplectomorphic to $B^{2n}(r)$. Similarly $Z \subseteq \mathbb{R}^{2n}$ is a **linear** (resp. **affine**) **symplectic cylinder** of radius R if it is linearly (resp. affinely symplectomorphic to $Z^{2n}(R)$.

The radius of a symplectic ball or cylinder is invariant. This follows from the above theorem: if B is symplectomorphic to $B^{2n}(r)$ and $B^{2n}(r')$, then $B^{2n}(r), B^{2n}(r')$ are symplectomorphic. By the above theorem this means that $r = r'$. And the radius of a symplectic cylinder is

$$R = \sup\left\{r > 0 \;\middle|\; \text{there exists an affine/linear symplectomorphism } \psi \text{ such that } \psi(B^{2n}(r)) \subseteq Z^{2n}(R)\right\}.$$

A matrix $\Psi \in \mathrm{Mat}_{2n}$ has the **linear nonsqueezing property** if for every linear symplectic ball B of radius r and linear symplectic cylinder Z of radius R

$$\Psi B \subseteq Z \Longrightarrow r \leq R.$$

By The affine nonsqueezing theorem any symplectic matrix has the linear nonsqueezing property. Call a matrix **anti-symplectic** if $\Psi^*\omega_0 = -\omega_0$. The affine nonsqueezing theorem clearly holds for anti-symplectic matrices as well.

Theorem 1.4.2 (The affine rigidity theorem)

Let $\Psi \in \mathrm{Mat}_{2n}$ be nonsingular such that Ψ, Ψ^{-1} have the linear nonsqueezing property. Then Ψ is either symplectic or anti-symplectic.

Proof

Suppose Ψ is neither symplectic nor anti-symplectic. Then neither is $\Psi^\top$ so there are $u, v \in \mathbb{R}^{2n}$ such that

$$\omega_0(\Psi^\top u, \Psi^\top v) \neq \pm\omega_0(u,v).$$

We may assume that neither side is zero, and potentially by replacing Ψ with Ψ^{-1}, $\left|\omega_0(\Psi^\top u, \Psi^\top v)\right| < |\omega_0(u,v)|$. We can rescale u so that $\omega_0(u,v) = 1$ so that

$$0 < \lambda^2 = \left|\omega_0(\Psi^\top u, \Psi^\top v)\right| < \omega_0(u,v) = 1.$$

Thus $\mathrm{span}\,\{u,v\}$ is a symplectic subspace of which u, v is a symplectic basis, so we can extend it to a symplectic basis of all of $\mathbb{R}^{2n}$ by considering its symplectic complement. So let $u_1, v_1, \ldots, u_n, v_n$ be a symplectic basis of $\mathbb{R}^{2n}$ with $u = u_1, v = v_1$. Now let $u_1' = \lambda^{-1}\Psi^\top u$ and $v_1' = \pm\lambda^{-1}\Psi^\top v$ where the sign is so that $\omega_0(u_1', v_1') = 1$. Extend this to a symplectic basis $u_1', v_1', \ldots, u_n', v_n'$ of $\mathbb{R}^{2n}$.

Define $\Phi, \Phi' \in \mathrm{Sp}(2n)$ to be the symplectic matrices which map the standard basis $e_1, \ldots, f_n$ to $u_1, \ldots, v_n$ and $u_1', \ldots, v_n'$ respectively. Then $A = (\Phi')^{-1}\Psi^\top\Phi$ satisfies

$$Ae_1 = \lambda e_1, \qquad Af_1 = \pm\lambda f_1.$$

Thus $A^\top$ maps the unit ball $B^{2n}(1)$ in $Z^{2n}(\lambda)$: if $|z| \leq 1$ then

$$\left\langle A^\top z, e_1\right\rangle^2 + \left\langle A^\top z, f_1\right\rangle^2 = \langle z, Ae_1\rangle^2 + \langle z, Af_1\rangle^2 = \lambda^2(\langle z, e_1\rangle^2 + \langle z, f_1\rangle^2) \leq \lambda^2.$$

Because $\lambda < 1$, this means that $A^\top$ does not have the linear squeezing property. Because Φ, Φ' are symplectic this means that Ψ does not have the linear squeezing property. ■

Let $A \subseteq \mathbb{R}^{2n}$ be an arbitrary subset. Define its **linear symplectic width** to be

$$w_L(A) = \sup\left\{\pi r^2 \;\middle|\; \psi(B^{2n}(r)) \subseteq A \text{ for some } \psi \in \mathrm{ASp}(\mathbb{R}^{2n})\right\}.$$

The linear symplectic width has the following properties:

(1) **monotonicity**: if $\psi(A) \subseteq B$ for $\psi \in \mathrm{ASp}(\mathbb{R}^{2n})$ then $w_L(A) \leq w_L(B)$.

(2) **conformality**: $w_L(\lambda A) = \lambda^2 w_L(A)$.

(3) **nontriviality**: $w_L(B^{2n}(r)) = w_L(Z^{2n}(r)) = \pi r^2$.

(1) is obvious, (2) is because $\psi(B^{2n}(r)) \subseteq A$ iff $\psi(B^{2n}(\lambda r)) \subseteq \lambda A$. (3) is due to The affine nonsqueezing theorem: $B^{2n}(r')$ embeds into $B^{2n}(r), Z^{2n}(r)$ iff $r' \leq r$ so $w_L(B^{2n}(r)) = \pi r^2$.

Lemma 1.4.3

Anti-symplectic linear maps have determinant $(-1)^n$. Furthermore anti-symplectic affine maps preserve linear symplectic width.

Proof

If $\Psi^*\omega_0 = -\omega_0$ then $\Psi^*\omega_0^n = (-1)^n\omega_0$. Because ω_0^n is a volume form, this means that $\det\Psi = (-1)^n$ as desired.

For the second point note that the composition of two anti-symplectic maps is symplectic, of a symplectic with an anti-symplectic is anti-symplectic. Define $D = \mathrm{diag}(I, -I)$; this is anti-symplectic and $D^2 = I$. Thus Ψ is anti-symplectic iff ΨD is symplectic. Furthermore D is orthonormal and thus $D(B^{2n}(r)) = B^{2n}(r)$.

This shows that $w_L(A)$ can be defined by taking the supremum over $\psi(B^{2n}(r)) \subseteq A$ where ψ is an anti-symplectic affine map. Thus w_L is monotonic with respect to anti-symplectic affine maps, and we have shown the second claim. ■

An **ellipsoid** is a subset $E \subseteq \mathbb{R}^{2n}$ of the form $\left\{z \in \mathbb{R}^{2n} \;\middle|\; Q(z) \leq 1\right\}$ where Q is any positive definite quadratic form. That is $Q(z) = z^\top \overline{Q} z$ for a symmetric positive-definite $\overline{Q} \in \mathrm{Sym}_+(2n)$. Notice that if $\Psi \in \mathrm{Mat}_{2n}$ is nonsingular then ΨE is an ellipsoid as well:

$$\Psi(E) = \left\{z \in \mathbb{R}^{2n} \;\middle|\; Q(\Psi^{-1}z) \leq 1\right\},$$

and $Q \circ \Psi^{-1}$ is a positive definite quadratic form.

Theorem 1.4.4

Let $\Psi \in \mathrm{Mat}_{2n}$ be a linear map. Then the following are equivalent.

(1) Ψ preserves the linear symplectic width of all ellipsoids.

(2) Ψ is either symplectic or anti-symplectic.

Proof

As shown above, symplectic and anti-symplectic maps preserve all widths, so (2) implies (1). So assume (2) and we will show that Ψ has the linear nonsqueezing property. Let B be a linear symplectic ball of radius r and Z a linear symplectic cylinder of radius R such that $\Psi B \subseteq Z$. Then by hypothesis

$$\pi r^2 = w_L(B) = w_L(\Psi B) \leq w_L(Z) = \pi R^2 \Longrightarrow r \leq R$$

as desired.

A symplectic ball cannot be embedded into a subspace of dimension $< 2n$, so if Ψ were singular then $w_L(\Psi B^{2n}(1)) = 0$, contradicting (1). Thus Ψ is nonsingular, and Ψ^{-1} satisfies (1) because

$$w_L(\Psi^{-1}E) = w_L(\Psi\Psi^{-1}E) = w_L(E)$$

as $\Psi^{-1}E$ is an ellipsoid.

Thus both Ψ and Ψ^{-1} have the linear nonsqueezing property, so by The affine rigidity theorem Ψ is symplectic or anti-symplectic. ■

Lemma 1.4.5

Let $g: V \times V \to \mathbb{R}$ be an inner product on a symplectic space. Then there exists a basis $u_1, \dots, u_n, v_1, \dots, v_n$ of V which is both g-orthogonal and ω-standard. More so, the basis can be chosen so that $g(u_j, u_j) = g(v_j, v_j)$ for all j.

Proof

Consider $V = \mathbb{R}^{2n}$ with the Euclidean inner product and assume that

$$\omega(z, w) = \langle z, Aw \rangle .$$

Then A is nondegenerate and antisymmetric: $A^\top = -A$. Thus $iA \in \mathrm{Mat}_{2n}(\mathbb{C})$ is Hermitian, which means its eigenvalues are all real, so A's eigenvalues are all pure imaginary. Furthermore the eigenvalues come in pairs $\pm i\alpha_j$ for $j = 1, \dots, n$, with $\alpha_j > 0$.

As iA is Hermitian, we can find $z_j = u_j + iv_j \in \mathbb{C}^{2n}$ an orthonormal basis of eigenvectors:

$$Az_j = i\alpha_j z_j, \qquad \overline{z_j}^\top z_k = \delta_{jk}.$$

And $\overline{z}_j$ fill out the basis of eigenvectors, so we have that $z_j^\top z_k = 0$. Thus

$$Au_j = -\alpha_j v_j, \qquad Av_j = \alpha_j u_j, \qquad u_j^\top v_k = u_j^\top u_k = v_j^\top v_k = 0 \text{ for } j \neq k.$$

And so

$$\omega(u_j, v_j) = u_j^\top A v_j = \alpha_j |u_j|^2 > 0,$$

and similarly $\omega(u_j, v_k) = \omega(u_j, u_k) = \omega(v_j, v_k) = 0$ for $j \neq k$. We can thus rescale u_j, v_k to get the desired basis.

Let $r = (r_1, \dots, r_n)$ be an n-tuple with $0 < r_1 \leq \cdots \leq r_n$, and define the ellipsoid

$$E(r) = \left\{ z \in \mathbb{C}^n \;\middle|\; \sum_{i=1}^n \left| \frac{z_i}{r_i} \right|^2 \leq 1 \right\}.$$

Lemma 1.4.6

Any ellipsoid

$$E = \left\{ w \in \mathbb{R}^{2n} \;\middle|\; w^\top A w \leq 1 \right\}$$

is symplectomorphic to a unique $E(r)$.

Proof

We can write E as $\left\{w \in \mathbb{R}^{2n} \;\middle|\; g(w,w) \leq 1\right\}$ for the inner product

$$g(u,v) = u^\top A v.$$

By the above lemma there is a basis $u_1, \ldots, u_n, v_1, \ldots, v_n$ of $\mathbb{R}^{2n}$ which is both symplectic and orthogonal for g. Define r_j so that $g(u_j, u_j) = g(v_j, v_j) = 1/r_j^2$. We may rearrange the u_j, v_j so that the r_j are ascending; this does not affect them forming a symplectic basis.

Let $\Phi\colon \mathbb{R}^{2n} \to \mathbb{R}^{2n}$ be the symplectomorphism mapping the standard basis to this new basis:

$$\Phi(z) = \sum_{j=1}^{n} (x_j u_j + y_j v_j), \qquad z = (x_1, \ldots, x_n, y_1, \ldots, y_n).$$

Thus

$$g(\Psi z, \Psi z) = \sum_{j=1}^{n} \frac{x_j^2 + y_j^2}{r_j^2},$$

which means that

$$\Psi^{-1} E = E(r),$$

as desired.

Now to show uniqueness of $r = (r_1, \ldots, r_n)$, define the matrix

$$\Delta(r) = \operatorname{diag}(1/r_1^2, \ldots, 1/r_n^2, 1/r_1^2, \ldots, 1/r_n^2)$$

so that

$$E(r) = \left\{z \in \mathbb{R}^{2n} \;\middle|\; z^\top \Delta(r) z \leq 1\right\}.$$

Because $\Psi E(r)$ maps to the ellipsoid defined by $\Psi^\top \Delta(r) \Psi$, we want to show that if $\Psi^\top \Delta(r) \Psi = \Delta(r')$, then $r = r'$. We know that $J_0 \Psi^\top = \Psi^{-1} J_0$ so

$$\Psi^{-1} J_0 \Delta(r) \Psi = J_0 \Delta(r').$$

And so $J_0\Delta(r)$ and $J_0\Delta(r')$ have the same eigenvalues. But these matrices correspond to the complex matrices $i \operatorname{diag}(1/r_1^2, \ldots, 1/r_n^2)$ and $i \operatorname{diag}(1/{r'_1}^2, \ldots, 1/{r'_n}^2)$. Thus we must have $r = r'$ as desired. ■

Note that $E(r, \ldots, r)$ is just $B^{2n}(r)$.

Note:
In the case of $n = 1$, symplectic forms are volume forms and symplectomorphisms are the same as area-preserving maps. Thus the above lemma states that every ellipse in $\mathbb{R}^2$ can be mapped into a circle by an area-preserving linear transformation, and that this circle is unique.

Define the **symplectic spectrum** of an ellipsoid E to be the unique n-tuple $r = (r_1, \ldots, r_n)$ with $0 < r_1 \leq \cdots \leq r_n$ such that E is symplectomorphic to $E(r)$. This is an invariant which uniquely determines the ellipsoid up to symplectomorphism. So if E has symplectic spectrum r, we see that

$$\operatorname{vol} E = \operatorname{vol} E(r) = \frac{\pi^n}{n!} \prod_{j=1}^{n} r_j^2$$

where the last equality is just an explicit integral.

Theorem 1.4.7
Let $E \subseteq \mathbb{R}^{2n}$ be an ellipsoid, then

$$w_L(E) = \sup_{B \subseteq E} w_L(B) = \inf_{Z \supseteq E} w_L(E),$$

where the supremum runs over all linear (or affine) symplectic balls contained in E, and the infimum over all linear symplectic cylinders containing E. Let $r = (r_1, \dots, r_n)$ be E's symplectic spectrum, then

$$w_L(E) = \pi r_1^2.$$

Proof

Suppose $\Psi \in \mathrm{Sp}(2n)$ is a symplectomorphism such that $\Psi E = E(r)$. Then we know

$$B^{2n}(r_1) \subseteq E(r) \subseteq Z^{2n}(r_1) \Longrightarrow \Psi^{-1}B^{2n}(r_1) \subseteq E \subseteq \Psi^{-1}Z^{2n}(r_1).$$

The widths of the bounds are both πr_1^2 so we have

$$w_L(E) = \pi r_1^2.$$

And furthermore we then get

$$\inf_{Z \supseteq E} w_L(Z) \leq \pi r_1^2 \leq \sup_{B \subseteq E} w_L(B)$$

because $\Psi^{-1}Z^{2n}(r_1)$ is in the infimum on the left, and $\Psi^{-1}B^{2n}(r_1)$ is in the supremum on the right. But if $B \subseteq E$ is a symplectic ball and $E \subseteq Z$ is a symplectic cylinder, then $w_L(B) \leq w_L(Z)$ by monotonicity, and so every element in the supremum is less than every element in the infimum, so we have equality. ■

For an ellipsoid E, define its **dual** to be

$$E^* = \left\{ v \in \mathbb{R}^{2n} \;\middle|\; \langle v, e \rangle \leq 1 \text{ for all } e \in E \right\}.$$

Let us temporarily denote the ellipsoid determined by the positive definite quadratic form A by $E(A)$, and the inner product $g_A(v,u) = v^\top A u$. Then we claim $E(A)^* = E(A^{-1})$. Notice that $v \in E(A^{-1})$ iff $v^\top A^{-1} v \leq 1$ iff $A^{-1}v \in E(A)$. Thus for all $e \in E(A)$ we have by Cauchy-Schwarz

$$\langle v, e \rangle = g_A(A^{-1}v, e) \leq 1$$

so $v \in E(A)^*$. Conversely if $v \in E(A)^*$ then

$$\langle v, e \rangle = g_A(A^{-1}v, e) \leq 1$$

for all $e \in E(A)$. Let $\alpha = \sqrt{g_A(A^{-1}v, A^{-1}v)}$ be $A^{-1}v$'s norm, so that $A^{-1}v/\alpha$ has g_A-norm 1 and thus is in $E(A)$. But

$$g_A(A^{-1}v, A^{-1}v/\alpha) = \alpha \leq 1$$

meaning $A^{-1}v \in E(A)$ and thus $v \in E(A^{-1})$ as desired.

Note that A being positive definite means A^{-1} is too, so $E(A^{-1}) = E(A)^*$ is an ellipsoid. Furthermore it is immediate from this that $E^{**} = E$ and $(\Psi E)^* = (\Psi^\top)^{-1}E^*$, the latter being because

$$(\Psi E(A))^* = E((\Psi^\top)^{-1}A\Psi^{-1})^* = E(\Psi A^{-1}\Psi^\top) = (\Psi^\top)^{-1}E(A)^*.$$

1.5 Linear complex structures

Definition 1.5.1

A **(linear) complex structure** on a real vector space is an automorphism

$$J\colon V \to V \qquad \text{such that} \qquad J^2 = -I.$$

Denote the space of linear complex structures over V by

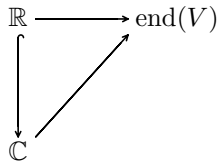
$$\mathcal{J}(V) \subseteq \operatorname{end}(V).$$

The archetypal linear complex structure is

$$J_0 = \begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix}$$

over $\mathbb{R}^{2n}$.

V having a complex structure is equivalent to it having the structure of a $\mathbb{C}$-vector space compatible with its real vector space structure. This is equivalent to the existence of a ring map $\mathbb{C} \to \operatorname{end}(V)$ which is a lifting of the ring map $\mathbb{R} \to \operatorname{end}(V)$ which defines V's real linear structure:



That is, the existence of a map of $\mathbb{R}$-algebras $\mathbb{C} \to \operatorname{end}(V)$. Because $\mathbb{C}$ is generated by i as a real algebra, this is equivalent to the existence of $J \in \operatorname{end}(V)$ such that $J^2 = -I$. Spelling this out, if V has a complex structure then it becomes a complex vector space with multiplication

$$\mathbb{C} \times V \to V, \qquad (a + ib, v) \mapsto av + bJv.$$

Notice that this necessitates that V has even real dimension. This can be seen because

$$\dim_{\mathbb{R}} V = \dim_{\mathbb{C}} V \cdot \dim_{\mathbb{R}} \mathbb{C} = 2 \dim_{\mathbb{C}} V$$

and because $J^2 = -I$ means that $\det J^2 = (-1)^{\dim V}$ is positive so $\dim V$ is even.

Proposition 1.5.2

Let V be a $2n$-dimensional real vector space with complex structure J. Then there is a vector space isomorphism $\Phi\colon \mathbb{R}^{2n} \to V$ such that $J\Phi = \Phi J_0$.

Proof

Viewing V as a complex vector space, there is a basis $v_1, \ldots, v_n$. Then $v_1, \ldots, v_n, Jv_1, \ldots, Jv_n$ is a basis of the real vector space V. Mapping the standard basis $e_1, \ldots, e_n, f_1, \ldots, f_n$ to this basis gives the desired result. ∎

Note that the map $\Phi\colon \operatorname{end}(V) \to \operatorname{end}(V)$ given by $A \mapsto A^2$ is smooth, and its differential at A is $X \mapsto AX + XA$. For $A \in \mathcal{J}(V)$ we have that

$$\ker d\Phi_A = \{X \in \operatorname{end}(V) \mid AX + XA = 0\},$$

which we claim has dimension $2n^2$. We can take $V = \mathbb{R}^{2n}$ and since there is an automorphism mapping A to J_0, we may assume $A = J_0$. Then

$$\begin{pmatrix} X & Y \\ Z & W \end{pmatrix}\begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix} + \begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix}\begin{pmatrix} X & Y \\ Z & W \end{pmatrix} = \begin{pmatrix} Y & -X \\ W & -Z \end{pmatrix} + \begin{pmatrix} -Z & -W \\ X & Y \end{pmatrix} = \begin{pmatrix} Y - Z & -X - W \\ X + W & Y - Z \end{pmatrix}.$$

And this is zero iff $Z = Y$ and $W = -X$, so we have $2n^2$ degrees of freedom as desired.

Thus Φ has constant rank $(2n)^2 - 2n^2 = 2n^2$ on $\mathcal{J}(V) = \Phi^{-1}(-I)$, which means that $\mathcal{J}(V)$ is an embedded submanifold of $\operatorname{end}(V)$ of dimension $2n^2$.

A linear complex structure J on V induces an orientation as follows. Let $v_1, \dots, v_n \in V$ form a complex basis so that $v_1, Jv_1, \dots, v_n, Jv_n$ forms a real basis. This real basis is the orientation. This orientation is well-defined, as in it is not dependent on v_i. Let $u_1, \dots, u_n$ be another complex basis, and let Φ be the complex map $v_i \mapsto u_i$. Let us write

$$u_i = \sum_j a_{ij} v_j + b_{ij} J v_j, \qquad Ju_i = \sum_j a_{ij} J v_j - b_{ij} v_j$$

so that the representation of the base transfer matrix (of $v_1, \dots, v_n, Jv_1, \dots, Jv_n \to u_1, \dots, u_n, Ju_1, \dots, Ju_n$) is

$$\begin{pmatrix} A & -B \\ B & A \end{pmatrix} \leftrightarrow A + iB,$$

and we know that this has determinant $|\det(A+iB)|^2 = |\det \Phi|^2 > 0$, so they are compatibly oriented. Note here that we permuted the oriented basis, but we did this for both using the same permutation, so the transfer matrix has the same determinant.

A linear complex structure is **compatible** with an orientation if it induces that orientation. For example J_0 induces the standard orientation on $\mathbb{R}^{2n}$.

Proposition 1.5.3

(1) $\mathcal{J}(\mathbb{R}^{2n})$ is diffeomorphic to $\mathrm{GL}_{2n}(\mathbb{R})/\mathrm{GL}_n(\mathbb{C})$ and thus has two connected components.

(2) The connected component $\mathcal{J}^+(\mathbb{R}^{2n})$ containing J_0 is the space of all complex structures which are compatible with the standard orientation. It is thus diffeomorphic to $\mathrm{GL}^+_{2n}(\mathbb{R})/\mathrm{GL}_n(\mathbb{C})$ and is homotopy equivalent to

$$\mathcal{J}^+(\mathbb{R}^{2n}) \cap \mathrm{SO}(2n) \cong \frac{\mathrm{SO}(2n)}{\mathrm{U}(n)}$$

of orthogonal complex structures which are compatible with the standard orientation.

(3) $\mathcal{J}^+(\mathbb{R}^4)$ is homotopy equivalent to S^2.

Proof

Consider the smooth map

$$\mathrm{GL}_{2n}(\mathbb{R}) \to \mathcal{J}(\mathbb{R}^{2n}), \qquad \Psi \mapsto \Psi J_0 \Psi^{-1}.$$

By the above lemma this is a surjection, and it quotients to a bijective smooth $\mathrm{GL}_{2n}(\mathbb{R})/\mathrm{GL}_n(\mathbb{C}) \to \mathcal{J}(\mathbb{R}^{2n})$. Both of these manifolds have dimension $2n^2$ and the above map is a submersion, so its quotient is a submersion as well, thus the quotient is a bijective submersion between manifolds of the same dimension, and thus a diffeomorphism.

The connected components of $\mathrm{GL}_{2n}(\mathbb{R})/\mathrm{GL}_n(\mathbb{C})$ correspond to the connected components of $\mathrm{GL}_{2n}(\mathbb{R})$, which are determined by the sign of the determinant, equivalently the oriented and non-oriented matrices. As $\det J = 1$, it lies in $\mathrm{GL}^+_{2n}(\mathbb{R})/\mathrm{GL}_n(\mathbb{C})$.

Now $\mathrm{O}(2n)/\mathrm{U}(n) \hookrightarrow \mathrm{GL}_{2n}(\mathbb{R})/\mathrm{GL}_n(\mathbb{C})$ is a strong deformation retract, via the homotopy

$$H\colon [0,1] \times \mathrm{GL}_{2n}(\mathbb{R}) \to \mathrm{GL}_{2n}(\mathbb{R}), \qquad H(t,\Psi) = \Psi(\Psi^\top \Psi)^{-t/2}.$$

This homotopy preserves the quotients, so it descends into them giving the desired homotopy. The connected component $\mathrm{GL}^+_{2n}(\mathbb{R})/\mathrm{GL}_n(\mathbb{C})$ corresponds to $\mathrm{SO}(2n)/\mathrm{U}(n) \cong \mathcal{J}^+(\mathbb{R}^{2n}) \cap \mathrm{SO}(2n)$.

For (3) let $n = 2$ and let e_0, e_1, e_2, e_3 be the standard basis of $\mathbb{R}^4$ and denote a vector by $x = (x_0, x_1, x_2, x_3)$. Then a matrix

$$J \in \mathcal{J}(\mathbb{R}^4) \cap \mathrm{SO}(4)$$

is determined by Je_0. And J satisfies $\langle v, Jv \rangle = 0$ because J is orthogonal and similar to J_0, they are orthogonally similar, and because this identity holds for J_0 it holds for J. Thus e_0, Je_0 form an orthonormal basis of a 2-plane $E \subseteq \mathbb{R}^4$.

Now for any unit $v \in E^\perp$, v, Jv form an orthonormal basis of $E^\perp$. And because J is compatible with the standard orientation, e_0, Je_0, v, Jv is positively oriented. There are only two unit vectors in $E^\perp$ orthogonal to v, and only one makes the basis oriented. So Jv is uniquely defined.

Thus $\mathcal{J}^+(\mathbb{R}^4) \cap \mathrm{SO}(2n) \to S^2$ given by $J \mapsto Je_0$ is a continuous bijection, and is seen to be a homeomorphism. Thus $\mathcal{J}^+(\mathbb{R}^4)$ is homotopy equivalent to S^2. ■

Definition 1.5.4

Let (V, ω) be a symplectic space. Then a complex structure $J \in \mathcal{J}(V)$ is **compatible** with ω if

$$J^*\omega = \omega, \qquad \omega(v, Jv) > 0 \text{ for all } v \in V.$$

Denote the space of compatible complex structures on (V, ω) by $\mathcal{J}(V, \omega)$.

If J is compatible with ω then

$$g_J(v, u) = \omega(v, Ju)$$

is an inner product on V, and J is skew-adjoint with respect to this inner product: $g_J(v, Ju) = -g_J(Jv, u)$.

Proposition 1.5.5

Let (V, ω) be a symplectic space and J a complex structure on V. The following are then equivalent.

(1) J is compatible with ω.

(2) (V, ω) has a symplectic basis of the form

$$v_1, \dots, v_n, Jv_1, \dots, Jv_n.$$

(3) There exists a vector space isomorphism $\Psi\colon \mathbb{R}^{2n} \to V$ such that

$$\Psi^*\omega = \omega_0, \qquad \Psi^* J = J_0.$$

(4) J satisfies $\omega(v, Jv) > 0$ and

$$\Lambda \in \mathcal{L}(V, \omega) \implies J\Lambda \in \mathcal{L}(V, \omega).$$

Proof

Suppose (1), and let $\Lambda \subseteq V$ be a Lagrangian subspace. Choose an orthonormal basis $v_1, \dots, v_n$ of Λ with respect to g_J, then

$$\omega(v_i, Jv_j) = g_J(v_i, v_j) = \delta_{ij}, \qquad \omega(Jv_i, Jv_j) = \omega(v_i, v_j) = 0$$

(remember that $v_i \in \Lambda$ and ω is trivial on Λ). Thus $v_1, \dots, v_n, Jv_1, \dots, Jv_n$ forms a symplectic basis as desired.

(2) then implies (3) by mapping

$$\Psi(z) = \sum_{i=1}^{n} (x_i v_i + y_i J v_i), \qquad z = (x_1, \dots, x_n, y_1, \dots, y_n).$$

(3) is easily seen to imply (1).

Now (1) implies (4), because $J\Lambda$ has the same dimension of Λ and for $v, u \in \Lambda$,

$$\omega(Jv, Ju) = \omega(u, v) = 0$$

so that $J\Lambda$ is isotropic and thus Lagrangian.

Now (4) implies (1): $g_J(v, u) = \omega(v, Ju)$ is positive definite by assumption, so we just need to show that it is symmetric. Suppose that g_J is not symmetric, then there are $v, u \in V$ such that

$$\omega(v, Ju) \neq \omega(u, Jv).$$

Then v is nonzero so $\omega(v, Jv) > 0$, and define

$$w = u - frac\omega(v, Ju)\omega(v, Jv)v.$$

Then

$$\omega(v, Jw) = 0, \qquad \omega(w, Jv) = \omega(u, Jv) - \omega(v, Ju) \neq 0.$$

Thus w, Jv are linearly independent and thus so too are Jw, v. Because $\omega(Jw, v) = 0$, we can extend v, Jw to a Lagrangian subspace Λ. Then $J\Lambda$ is Lagrangian, but $Jv, w \in J\Lambda$ have nonzero ω. ■

We now provide three proofs of the fact that $\mathcal{J}(V, \omega)$ is contractible. We do this in three different ways:

(1) The first method identifies $\mathcal{J}(V, \omega)$ with the space of symmetric, positive definite, and symplectic matrices. Then this is shown to be contractible.

(2) The second method constructs an explicit homotopy equivalence between $\mathcal{J}(V, \omega)$ and the space $\mathrm{Met}(V)$ of all inner products on V. This is based on proposition 1.5.7 which shows that $\mathcal{J}(V)$ is homotopy equivalent to the space of nondegenerate skew-symmetric bilinear forms on V. This method will show that every compact subgroup of $\mathrm{Sp}(2n)$ is conjugate to a subgroup of $\mathrm{U}(n)$.

(3) The third method identifies $\mathcal{J}(V, \omega)$ with the Siegel upper half space.

Proposition 1.5.6

Let (V, ω) be a $2n$-dimensional symplectic space. Then $\mathcal{J}(V, \omega)$ is diffeomorphic to $\mathrm{Sym}_+(2n) \cap \mathrm{Sp}(2n)$, which is contractible.

Proof

We may assume that $V = \mathbb{R}^{2n}$ and $\omega = \omega_0$ in lieu of the previous theorem. Then $J \in \mathcal{J}(\mathbb{R}^{2n}, \omega_0)$ iff

$$J^2 = -1, \qquad J^\top J_0 J = J_0, \qquad \langle v, -J_0 Jv\rangle = v^\top(-J_0 J)v > 0.$$

Thus

$$(J_0 J)^\top = -J^\top J_0 = -J_0 J^{-1} = J_0 J,$$

and so

$$P = -J_0 J$$

is symmetric, positive definite, and symplectic (because J_0, J are symplectic). Conversely if P has these properties then $J = J_0 P$ is a compatible complex structure. These maps are both smooth inverses, so we have the desired diffeomorphism.

And $\mathrm{Sym}_+(2n) \cap \mathrm{Sp}(2n)$ is contractible by the null-homotopy

$$H(t, P) = P^t,$$

which is continuous by lemma 1.2.5 and well-defined by lemma 1.2.6. This is a null homotopy to the point I as $H_0(\bullet) = I$. ■

For the second proof we show that $\mathcal{J}(V)$ is homotopy equivalent to the space of symplectic forms on V:

$$\mathcal{S}(V) = \left\{\omega \in \Lambda^2(V^*) \;\middle|\; \omega \text{ is nondegenerate}\right\}.$$

The homotopy equivalence is given explicitly, depending only on a choice of inner product on V. Define

$$\mathrm{Met}(V) = \left\{g \in \mathrm{Sym}^2(V^*) \;\middle|\; g \text{ is positive definite}\right\},$$

the space of all inner products on V. $\mathrm{Met}(V)$ is an open convex subspace of $\mathrm{Sym}^2(V^*)$: it is convex because the positive multiples and sums of inner products are inner products, it is open because $\mathrm{Sym}_+(n) \subseteq \mathrm{Sym}(n)$ is (a similar agument to the proof of lemma 1.2.5 shows that for a positive definite matrix A there is $0 < m$ and a neighborhood U in Sym such that all matrices in U have eigenvalues bound below by m, thus are positive definite). Because $\mathrm{Met}(V)$ is convex, it is contractible.

For $g \in \mathrm{Met}(V)$ define

$$\mathcal{J}(V,g) = \{J \in \mathcal{J}(V) \mid J^*g = g\}, \qquad \mathcal{S}(V,g) = \{g(J\bullet,\bullet) \mid J \in \mathcal{J}(V,g)\}.$$

Notice that $J^*g = g$ iff $\omega_g = g(J\bullet,\bullet) \in \mathcal{S}(V)$: if $J^*g = g$ then

$$\omega_g(v,u) = g(Jv,u) = g(-v,Ju) = -g(Ju,v) = -\omega_g(u,v),$$

and if $\omega_g \in \mathcal{S}(V)$ then

$$g(Ju,Jv) = \omega_g(u,Jv) = -\omega_g(Jv,u) = -g(-v,u) = g(v,u) = g(u,v).$$

And so $\mathcal{S}(V,g) \subseteq \mathcal{S}(V)$.

Elements of $\mathcal{J}(V,g)$ and $\mathcal{S}(V,g)$ are said to be **compatible with** g.

Now consider the map

$$\mathrm{Met}(V) \times \mathcal{J}(V) \to \mathcal{S}(V), \qquad (g,J) \mapsto \omega_{g,J} = \frac{1}{2}(g(J\bullet,\bullet) - g(\bullet,J\bullet)).$$

This is well-defined as $\omega_{g,J}$ is clearly antisymmetric bilinear, and it is nondegenerate

$$\omega_{g,J}(v,Jv) = \frac{1}{2}(g(Jv,Jv) + g(v,v)) > 0.$$

Furthermore this map is smooth as it can be seen as evaluating a polynomial at a matrix. Recall that $\mathrm{GL}(V)$ acts on $T^2(V^*)$ (bilinear forms on V) by $(\Psi^*\alpha)(v,u) = \alpha(\Psi^{-1}v, \Psi^{-1}u)$, and this is a smooth action which restricts to $\mathrm{Met}(V)$ and $\mathcal{S}(V)$. The map above is then $\mathrm{GL}(V)$-equivariant:

$$\begin{aligned}\omega_{\Psi^*g,\Psi^*J}(v,u) = \frac{1}{2}((\Psi^*g)(\Psi^*Jv,u) - (\Psi^*g)(v,\Psi^*Ju)) &= \frac{1}{2}(g(J\Psi^{-1}v,\Psi^{-1}u) - g(\Psi^{-1}v,J\Psi^{-1}u)) \\ &= \omega_{g,J}(\Psi^{-1}v,\Psi^{-1}u) = (\Psi^*\omega_{g,J})(v,u).\end{aligned}$$

For a fixed $g \in \mathrm{Met}(V)$ this map restricts to $f_g\colon \mathcal{J}(V) \to \mathcal{S}(V)$ by $f_g(J) = \omega_{g,J}$. And for $J \in \mathcal{J}(V,g)$ we have that $g(J\bullet,\bullet) \in \mathcal{S}(V,g)$ meaning $g(\bullet,J\bullet) = -g(J\bullet,\bullet)$ so that $f_g(J) = g(J\bullet,\bullet) \in \mathcal{S}(V,g)$. Thus $f_g\colon \mathcal{J}(V,g) \to \mathcal{S}(V,g)$ is a bijective $\mathrm{GL}(V)$-equivariant smooth map, and thus a diffeomorphism. And so we have the following commutative diagram

$$\begin{array}{ccc} \mathcal{J}(V,g) & \xrightarrow[\sim]{f_g} & \mathcal{S}(V,g) \\ \downarrow & & \downarrow \\ \mathcal{J}(V) & \xrightarrow{f_g} & \mathcal{S}(V) \end{array}$$

Proposition 1.5.7

Let V be an even-dimensional real vector space. Then there exists a $\mathrm{GL}(V)$-equivariant smooth map

$$\mathrm{Met}(V) \times \mathcal{S}(V) \to \mathcal{J}(V), \qquad (g, \omega) \mapsto J_{g,\omega}$$

with the following properties.

(1) For $g \in \mathrm{Met}(V), \omega \in \mathcal{S}(V), J \in \mathcal{J}(V)$, we have

$$J_{g,\omega} \in \mathcal{J}(V, \omega) \cap \mathcal{J}(V, g), \qquad J \in \mathcal{J}(V, \omega_{g,J})$$

and if $g = \omega(\bullet, J\bullet)$ then

$$J_{g,\omega} = J, \qquad \omega_{g,J} = \omega.$$

Thus for $g \in \mathrm{Met}(V)$, $\mathcal{J}(V, g) \to \mathcal{S}(V, g)$ mapping $J \mapsto \omega_{g,J}$ has inverse $\mathcal{S}(V, g) \to \mathcal{J}(V, g)$ mapping $\omega \mapsto J_{g,\omega}$.

(2) For every $g \in \mathrm{Met}(V)$, the map

$$\mathcal{S}(V) \to \mathcal{J}(V), \qquad \omega \mapsto J_{g,\omega}$$

is a homotopy equivalence with inverse

$$\mathcal{J}(V) \to \mathcal{S}(V), \qquad J \mapsto \omega_{g,J}.$$

(3) For every $g \in \mathrm{Met}(V)$, the inclusions $\mathcal{J}(V, g) \hookrightarrow \mathcal{J}(V)$ and $\mathcal{S}(V, g) \hookrightarrow \mathcal{S}(V)$ are homotopy equivalences with inverses

$$\begin{aligned} \mathcal{J}(V) &\to \mathcal{J}(V, g), \qquad J \mapsto J_{g,\omega_{g,J}} \\ \mathcal{S}(V) &\to \mathcal{S}(V, g), \qquad \omega \mapsto \omega_{g,J_{g,\omega}}. \end{aligned}$$

Proof

First we construct this map. For $g \in \mathrm{Met}(V)$ and $\omega \in \mathcal{S}(V)$ define $A_{g,\omega} \in \mathrm{GL}(V)$ by

$$g(A_{g,\omega} v, u) = \omega(v, u).$$

So $A_{g,\omega}$ is g-skew-adjoint and thus $-A_{g,\omega}^2$ is g-self-adjoint and g-positive definite. This means there exists a unique g-self-adjoint and g-positive definite automorphism $Q_{g,\omega}: V \to V$ such that $Q_{g,\omega}^2 = -A_{g,\omega}^2$ (consider the Euclidean case). By proposition 1.2.5, this is a smooth map. Note that $Q_{g,\omega}$ and $A_{g,\omega}$ commute (because $A_{g,\omega}^{-1} Q_{g,\omega} A_{g,\omega}$ is also g-self-adjoint and g-positive definite, so it is equal to $Q_{g,\omega}$).
So define

$$J_{g,\omega} = Q_{g,\omega}^{-1} A_{g,\omega} \in \mathcal{J}(V, \omega) \cap \mathcal{J}(V, g).$$

This is indeed a complex structure as

$$J_{g,\omega}^2 = A_{g,\omega}^2 Q_{g,\omega}^{-2} = -I,$$

and it is ω- and g-compatible as

$$g(J_{g,\omega}\bullet, J_{g,\omega}\bullet) = g(A_{g,\omega}\bullet, Q_{g,\omega}^{-2} A_{g,\omega}\bullet) = g(\bullet, \bullet)$$

and

$$\omega(J_{g,\omega}\bullet, J_{g,\omega}\bullet) = g(A_{g,\omega} J_{g,\omega}\bullet, J_{g,\omega}\bullet) = g(A_\omega \bullet, \bullet) = \omega(\bullet, \bullet),$$

finally

$$\omega(\bullet, J_{g,\omega}\bullet) = g(A_{g,\omega}\bullet, J_{g,\omega}\bullet) = -g(Q_{g,\omega}^{-1} A_{g,\omega}^2 \bullet, \bullet) > 0$$

as both $-A_{g,\omega}^2$ and $Q_{g,\omega}^{-1}$ are g-positive definite.

The map $(g,\omega)\mapsto J_{g,\omega}$ is smooth, we show that it is also $\mathrm{GL}(V)$-equivariant. For $\Psi\in\mathrm{GL}(V)$, first we see that $A_{\Psi^*g,\Psi^*\omega}=\Psi^*A_{g,\omega}=\Psi A_{g,\omega}\Psi^{-1}$:

$$(\Psi^*g)(\Psi A_{g,\omega}\Psi^{-1}v,u)=g(A_{g,\omega}\Psi^{-1}v,\Psi^{-1}u)=\omega(\Psi^{-1}v,\Psi^{-1}u)=(\Psi^*\omega)(v,u).$$

And $\Psi^*Q_{g,\omega}$ is Ψ^*g-self adjoint and positive definite and squaring gives $-\Psi^*A_{g,\omega}^2=-(\Psi^*A_{g,\omega})^2$. Thus

$$J_{\Psi^*g,\Psi^*\omega}=(\Psi^*Q_{g,\omega})^{-1}(\Psi^*A_{g,\omega})=\Psi^*(Q_{g,\omega}^{-1}A_{g,\omega})=\Psi^*J_{g,\omega}$$

as desired.

Now we show that this construction satisfies (1): we already showed that $J_{g,\omega}\in\mathcal{J}(V,\omega)\cap\mathcal{J}(V,g)$. And for $g=\omega(\bullet,J\bullet)$, we have that $A_{g,\omega}=J$ as

$$g(Jv,u)=g(u,Jv)=\omega(u,J^2v)=-\omega(u,v)=\omega(v,u),$$

then $Q=I$ is g-symmetric and positive definite and satisfies $Q^2=-A^2$, so $J_{g,\omega}=J$ as desired.

Now we need to show (2). For a set $g\in\mathrm{Met}(V)$, the composite

$$\psi\colon\mathcal{S}(V)\xrightarrow{\omega\mapsto J_{g,\omega}}\mathcal{J}(V)\xrightarrow{J\mapsto\omega_{g,J}}\mathcal{S}(V),\qquad\omega\mapsto\omega_{g,J_{g,\omega}}$$

is homotopic to the identity via the homotopy

$$\psi_\bullet\colon[0,1]\times\mathcal{S}(V)\to\mathcal{S}(V),\qquad\psi_t(\omega)=(1-t)\omega+t\omega_{g,J_{g,\omega}}.$$

This is well-defined as $J_{g,\omega}\in\mathcal{J}(V,\omega)\cap\mathcal{J}(V,\omega_{g,J_{g,\omega}})$, so it is compatible with both of these symplectic structures, thus $\psi_t(\omega)(v,J_{g,\omega}v)>0$ and so $\psi_t(\omega)$ is symplectic for all t. As $\psi_0=\mathrm{id}$ and $\psi_1=\psi$ we have the desired homotopy.

And consider the composite

$$\varphi\colon\mathcal{J}(V)\xrightarrow{J\mapsto\omega_{g,J}}\mathcal{S}(V)\xrightarrow{\omega\mapsto J_{g,\omega}}\mathcal{J}(V),\qquad J\mapsto J_{g,\omega_{g,J}}.$$

This homotopic to the identity via $\varphi_\bullet\colon[0,1]\times\mathcal{J}(V)\to\mathcal{J}(V)$ by

$$\varphi_t(J)=J_{g_t,\omega_{g_t,J}},\qquad g_t=\frac{1+t}{2}g+\frac{1-t}{2}J^*g.$$

Because $g_1=g$ we see that $\varphi_1(J)=J_{g,\omega_{g,J}}=\varphi(J)$. And $g_0=\frac{1}{2}(g+J^*g)$ is compatible with J (because $(-I)^*g=g$), and thus also with

$$\omega_0=\omega_{g_0,J}=\frac{1}{2}(g_0(J\bullet,\bullet)-g_0(\bullet,J\bullet))=\frac{1}{2}(g(J\bullet,\bullet)-g(J\bullet,\bullet))=\omega_{g,J}$$

so that

$$\psi_0(J)=J_{g_0,\omega_0}=J$$

as desired.

For (3), for $\omega\in\mathcal{S}(V,g)$ there is a $J\in\mathcal{J}(V)$ such that $g=\omega(\bullet,J\bullet)$, thus

$$J_{g,\omega}=J,\qquad\omega_{g,J_{g,\omega}}=\frac{1}{2}(g(J\bullet,\bullet)-g(\bullet,J\bullet))=\omega.$$

And so the composition $\mathcal{S}(V,g)\hookrightarrow\mathcal{S}(V)\to\mathcal{S}(V,g)$ is the identity. The other direction of the composition is ψ, which is homotopic to the identity as we already showed.

Similarly for $J\in\mathcal{J}(V,g)$, we have that

$$\omega_{g,J}=g(J\bullet,\bullet)\Longrightarrow g=\omega_{g,J}(\bullet,J\bullet)$$

and so

$$J_{g,\omega_{g,J}}=J$$

so that $\mathcal{J}(V,g)\hookrightarrow\mathcal{J}(V)\to\mathcal{J}(V,g)$ is the identity, and the other direction is φ, which is homotopic to the identity. ∎

Corollary 1.5.8

For every $\omega \in \mathcal{S}(V)$, there is a homotopy equivalence

$$\mathcal{J}(V,\omega) \underset{g \mapsto J_{g,\omega}}{\overset{J \mapsto g_J}{\rightleftarrows}} \mathrm{Met}(V), \qquad g_J = \omega(\bullet, J\bullet).$$

Thus $\mathcal{J}(V,\omega)$ is contractible.

Proof

By the above proposition, $J_{g_J,\omega} = J$ so one composition is the identity. The other composition is $\mathrm{Met}(V) \to \mathrm{Met}(V)$ given by $g \mapsto g_{J_{g,\omega}}$, which is homotopic to the identity because $\mathrm{Met}(V)$ is convex. ■

Note:
If $X \subseteq V$ is a convex subset of a topological vector space V, then every map $f\colon X \to X$ is homotopic to the identity via linear interpolation:

$$\varphi_t(x) = tf(x) + (1-t)x.$$

We now need to briefly explain Haar measures. Let V, W be normed real vector spaces of potentially infinite dimension. A linear map $F\colon V \to W$ is **bounded** if there is some real $C \geq 0$ such that

$$|F(v)|_W \leq C\,|v|_V \qquad \text{for all } v \in V.$$

This is equivalent to the following quantity being finite:

$$|F|_{\mathsf{op}} = \sup_{0 \neq v \in V} \frac{|Fv|_W}{|v|_V} = \sup_{|v|_V = 1} |Fv|_W < \infty.$$

This makes the space of bounded linear maps $\hom(V,W)$ into a normed vector space, as is easily verified.

The **Haar measure** on a compact topological group G is the unique bounded linear functional

$$C(G,\mathbb{R}) \to \mathbb{R}, \qquad f \mapsto \int_G f\,d\mu$$

from the space of continuous maps $C(G,\mathbb{R})$ (with the supremum norm), which is invariant under both left and right translations, and the Haar measure of the constant function 1 is 1. By invariance under left and right translations, we mean

$$\int_G f(\Phi\bullet)\,d\mu = \int_G f(\bullet\Phi)\,d\mu = \int_G f(\bullet)\,d\mu \qquad \text{for all } \Phi \in G.$$

The reason for calling this the Haar *measure* is because it can be, and commonly is, constructed as a measure μ over the Borel subsets of G which is unique in that it has certain properties, and the linear functional is just the Lebesgue integral of f relative to μ. Thus we can utilize results regarding the Lebesgue integral when discussing the Haar measure.

We will now prove the following proposition with the help of some lemmas.

Proposition 1.5.9

Every compact subgroup $G \subseteq \mathrm{Sp}(2n)$ is conjugate to a subgroup of $\mathrm{U}(n)$.

Lemma 1.5.10

There exists a symmetric positive definite $S \in \mathrm{Mat}_{2n}$ such that $\Psi^\top S \Psi = S$ for all $\Psi \in G$.

Proof

For all pairs of indices i, j, define the continuous linear functional $\pi_{i,j}\colon \mathrm{Mat}_{2n} \to \mathbb{R}$ mapping a matrix to its value at that coordinate, $A \mapsto A_{ij}$. Then define

$$S_{ij} = \int_{\Psi \in G} \pi_{ij}(\Psi^\top \Psi)\, d\mu.$$

Because $\Psi^\top \Psi$ is symmetric, we see immediately that S is. And S is positive definite: for $v = (v_1, \ldots, v_{2n})$ we have

$$v^\top S v = \sum_{i,j} v_i S_{ij} v_j = \sum_{i,j} \int_{\Psi \in G} v_i \pi_{ij}(\Psi^\top \Psi) v_j = \int_{\Psi \in G} v^\top \Psi^\top \Psi v.$$

Because $\Psi^\top \Psi$ is positive-definite, this is the integral of a positive function and is thus positive, so S is positive-definite.

A similar computation shows that for every $\Phi \in G$ we have

$$\Phi^\top S \Phi = \int_{\Psi \in G} \Phi^\top \Psi^\top \Psi \Phi \, d\mu = \int_{\Psi \in G} (\Psi\Phi)^\top (\Psi\Phi)\, d\mu = S$$

because of the left invariance of the Haar measure. ■

Lemma 1.5.11

There is an ω_0-compatible complex structure $J \in \mathcal{J}(\mathbb{R}^{2n}, \omega_0)$ which commutes with every element of G.

Proof

Let S be as in the above lemma, and let $g = \langle \bullet, S\bullet \rangle \in \mathrm{Met}(\mathbb{R}^{2n})$ be the inner product determined by S. Then both g and ω_0 are G-invariant, and thus so is $J_{g,\omega}$ from proposition 1.5.7: for $\Psi \in G$,

$$\Psi^* J_{g,\omega} = J_{\Phi^* g, \Phi^* \omega} = J_{g,\omega}.$$ ■

Proof (1.5.9)

Let J be as in the previous lemma, so there is a vector space isomorphism $\Phi\colon \mathbb{R}^{2n} \to \mathbb{R}^{2n}$ such that $\Phi^* \omega_0 = \omega_0$ and $\Phi^* J = J_0$. Thus $\Phi \in \mathrm{Sp}(2n)$ and $\Phi J \Phi^{-1} = J_0$. For every $\Psi \in G$, $\Phi\Psi\Phi^{-1} \in \mathrm{Sp}(2n)$ commutes with J_0 and thus is in $\mathrm{U}(n)$. So we have

$$\Phi G \Phi^{-1} \subseteq \mathrm{U}(n)$$

as desired. ■

Remark 1.5.12

Let V be even-dimensional and define

$$\mathcal{C}(V) = \{(\omega, J) \mid \omega \in \mathcal{S}(V),\ J \in \mathcal{J}(V, \omega)\},$$

the space of compatible pairs. The projections $\pi_{\mathcal{S}}\colon \mathcal{C}(V) \to \mathcal{S}(V)$ and $\pi_{\mathcal{J}}\colon \mathcal{C}(V) \to \mathcal{J}(V)$ are then homotopy

equivalences. Indeed for $g \in \mathrm{Met}(V)$, the maps

$$\begin{aligned} \iota_{\mathcal{S}} &: \mathcal{S}(V) \to \mathcal{C}(V), \qquad \iota_{\mathcal{S}}(\omega) = (\omega, J_{g,\omega}) \\ \iota_{\mathcal{J}} &: \mathcal{J}(V) \to \mathcal{C}(V), \qquad \iota_{\mathcal{J}}(J) = (\omega_{g,J}, J) \end{aligned}$$

are homotopy inverses of $\omega_{\mathcal{S}}, \omega_{\mathcal{J}}$ respectively. Indeed we see that $\pi_{\mathcal{S}} \circ \iota_{\mathcal{S}} = \mathrm{id}$ and $\pi_{\mathcal{J}} \circ \iota_{\mathcal{J}} = \mathrm{id}$. And $\iota_{\mathcal{S}} \circ \pi_{\mathcal{S}}$ maps (ω, J) to $(\omega, J_{g,\omega})$ which is homotopic to the identity via

$$[0,1] \times \mathcal{C}(V) \to \mathcal{C}(V), \qquad (t, J, \omega) \mapsto (\omega, J_{tg+(1-t)\omega(\bullet,J\bullet),\omega}),$$

and $\iota_{\mathcal{J}} \circ \pi_{\mathcal{J}}$ maps (ω, J) to $(\omega_{g,J}, J)$ which is homotopic to the identity via

$$[0,1] \times \mathcal{C}(V) \to \mathcal{C}(V), \qquad (t, J, \omega) \mapsto (t\omega_{g,J} + (1-t)\omega, J).$$

Remark 1.5.13

We can understand the above remark by focusing on the case $V = \mathbb{R}^{2n}$ with the standard symplectic and complex structures. We then get the following diagram

$$\begin{array}{ccccccccc} \frac{\mathrm{O}(2n)}{\mathrm{U}(n)} & \hookrightarrow & \frac{\mathrm{GL}_{2n}(\mathbb{R})}{\mathrm{GL}_n(\mathbb{C})} & \hookleftarrow & \frac{\mathrm{GL}_{2n}(\mathbb{R})}{\mathrm{U}(n)} & \to & \frac{\mathrm{GL}_{2n}(\mathbb{R})}{\mathrm{Sp}(2n)} & \hookleftarrow & \frac{\mathrm{O}(2n)}{\mathrm{U}(n)} \\ \downarrow \sim & & \downarrow \sim & & \downarrow \sim & & \downarrow \sim & & \downarrow \sim \\ \mathcal{J}(V,g) & \hookrightarrow & \mathcal{J}(V) & \xleftarrow{\pi_{\mathcal{J}}} & \mathcal{C}(V) & \xrightarrow{\pi_{\mathcal{S}}} & \mathcal{S}(V) & \hookleftarrow & \mathcal{S}(V,g) \end{array}$$

The horizontal arrows are the homotopy equivalences discussed thus far, and the vertical arrows are diffeomorphisms. The outer vertical arrows are obtained by quotienting $\Psi \mapsto \Psi^* J_0$ or $\Psi^* \omega_0$ and quotienting.

The inner vertical arrow is defined as follows. Recall that by proposition 1.5.5 $(\omega, J) \in \mathcal{C}(V)$ iff there is a linear isomorphism $\Psi : \mathbb{R}^{2n} \to V$ such that $\Psi^* \omega_0 = \omega_0$ and $\Psi J_0 \Psi^{-1} = J$. So define the surjection

$$\mathrm{GL}_{2n}(\mathbb{R}) \to \mathcal{C}(\mathbb{R}^{2n}), \qquad \Psi \mapsto \Psi^*(\omega_0, J_0) = (\Psi^* \omega_0, \Psi J_0 \Psi^{-1}).$$

The kernel is $\mathrm{Sp}(2n) \cap \mathrm{GL}_n(\mathbb{C}) = \mathrm{U}(n)$, giving the desired diffeomorphism.

Definition 1.5.14

Let (V, ω) be a symplectic space. Then a complex linear structure $J \in \mathcal{J}(V)$ is called **ω-tame** if

$$\omega(v, Jv) > 0 \text{ for all } v \neq 0.$$

Denote the space of ω-tame complex structures by $\mathcal{J}_\tau(V, \omega)$.

Note that $\mathcal{J}_\tau(V, \omega) \subseteq \mathcal{J}(V, \omega)$ is open, and for $J \in \mathcal{J}_\tau(V, \omega)$,

$$g_{\omega,J}(v, u) = \frac{1}{2}(\omega(v, Ju) - \omega(Jv, u))$$

is an inner product.

Proposition 1.5.15

The space $\mathcal{J}_\tau(V,\omega)$ is contractible.

Proof

Define the space of tame pairs

$$\mathcal{C}_\tau(V) = \{(\omega, J) \mid \omega \in \mathcal{S}(V),\ J \in \mathcal{J}_\tau(V,\omega)\}.$$

Now for $(\omega, J) \in \mathcal{C}_\tau(V)$ define the 2-form

$$\pi_1(\omega, J) = \frac{1}{2}(\omega + J^*\omega).$$

Then

$$\frac{1}{2}(\omega + J^*\omega)(\bullet, J\bullet) = \frac{1}{2}(\omega(\bullet, J\bullet) - \omega(J\bullet, \bullet)) = g_{\omega,J},$$

which means that $\pi_1(\omega, J)$ is non-degenerate and thus symplectic. Furthermore $\pi_1(\omega, J)$ is compatible with J and defines the same inner product as ω, $g_{\omega,J}$.

The fiber of the projection

$$\mathrm{proj}\colon \mathcal{C}_\tau(V) \to \mathcal{C}(V), \qquad (\omega, J) \mapsto \left(\frac{1}{2}(\omega + J^*\omega), J\right)$$

over $(\omega, J) \in \mathcal{C}(V)$ consists of all $(\omega + \tau, J) \in \mathcal{C}_\tau(V)$ for $\tau \in \Lambda^2(V^*)$ such that $\tau + J^*\tau = 0$. Any such pair is tame, as

$$\tau(v, Jv) = -\tau(Jv, J^2v) = \tau(v, Jv) \Longrightarrow \tau(v, Jv) = 0$$

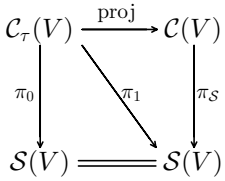
and thus $(\omega + \tau)(v, Jv) > 0$ as ω is compatible.

Note that the set of all 2-forms τ such that $\tau + J^*\tau = 0$ is a vector space. Thus we see that $\mathrm{proj}\colon \mathcal{C}_\tau(V) \to \mathcal{C}(V)$ is a vector bundle, and thus is a homotopy equivalence. By remark 1.5.12 so is $\pi_{\mathcal{S}}\colon \mathcal{C}(V) \to \mathcal{S}(V)$, and their composition $\pi_{\mathcal{S}} \circ \mathrm{proj}\colon \mathcal{C}_\tau(V) \to \mathcal{S}(V)$ is the map π_1.

Define

$$\pi_0\colon \mathcal{C}_\tau(V) \to \mathcal{S}(V), \qquad \pi_0(\omega, J) = \omega$$

and consider the diagram



The top triangle commutes, but the bottom does not. Despite this, π_1 and π_0 are homotopic via $\pi_t = (1-t)\pi_0 + t\pi_1$: because both $\pi_0(\omega, J)$ and $\pi_1(\omega, J)$ are compatible with J, their convex combinations are symplectic. Thus $\pi_0\colon \mathcal{C}_\tau(V) \to \mathcal{S}(V)$ is a homotopy equivalence, and as it is a fibration, its fibers $\pi_0^{-1}\omega \cong \mathcal{J}_\tau(V,\omega)$ are contractible. ■